Supplementary Methods for

Physics-guided discriminability of process-level observables in partially observable cancer systems

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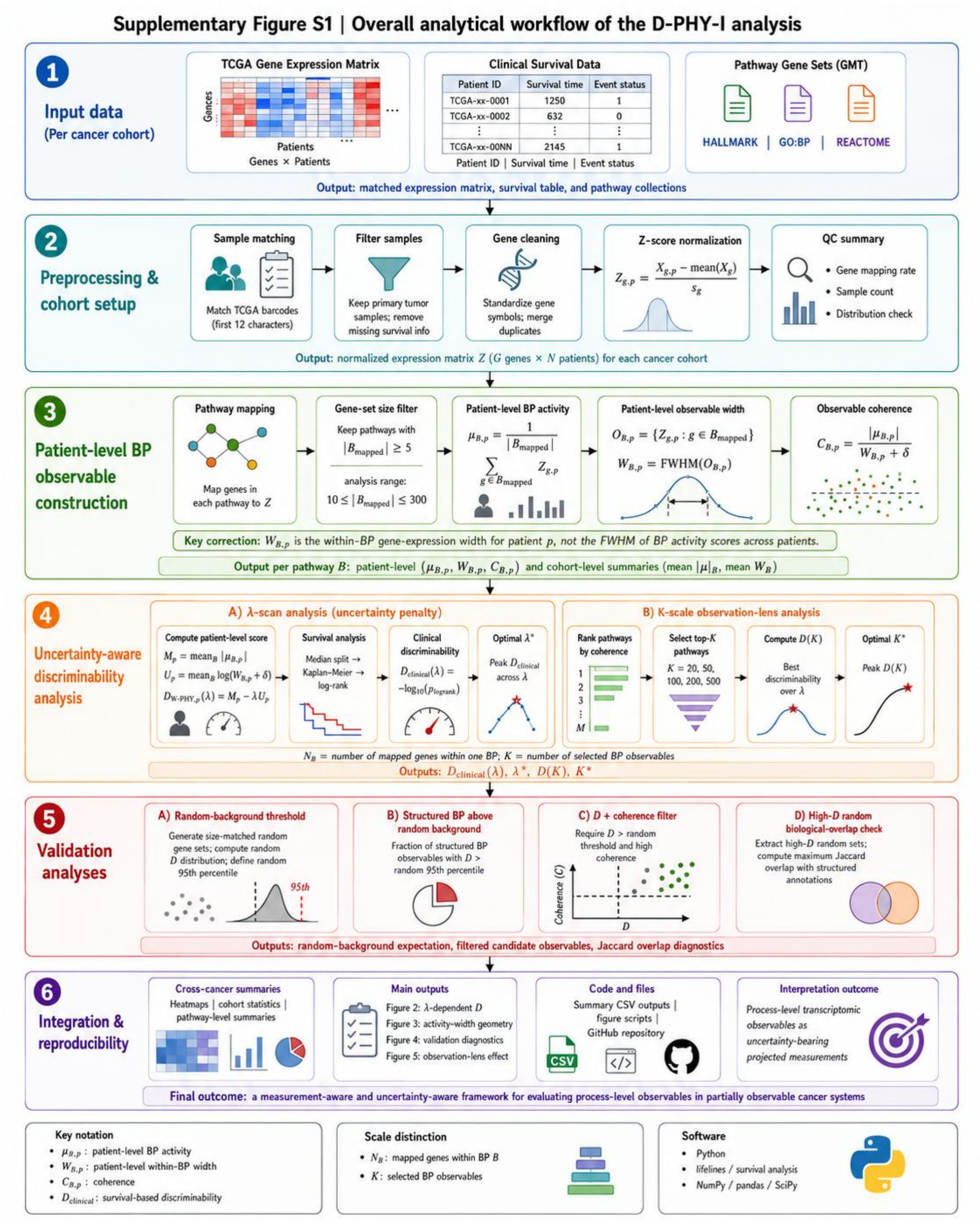
Supplementary Methods organization

The Supplementary Methods are organized to support the analytical sequence used in the main manuscript. SM1–SM3 describe data sources, TCGA cohort preparation, transcriptomic preprocessing, and biological-process gene-set mapping. SM4–SM6 define the core activity–width representation, including patient-level BP activity $\mu_{B,p}$, patient-level observable width $W_{B,p}$, cohort-level activity–width summaries, coherence, and the distinction between mapped gene-set size N_B and observable-set scale K. SM7–SM9 describe the uncertainty-aware patient-level score, survival-based clinical discriminability, and λ-scan analysis. SM10–SM11 define the random-baseline workflow, D + coherence filtering, and biological-overlap analysis used to support the Figure 4 diagnostic framework. SM12 describes the *K*-scale observation-lens analysis used for Figure 5. SM13 summarizes supplementary figure generation, code organization, data availability, and reproducibility safeguards.

Symbol	Meaning	Used in
B	Biological-process or pathway gene set	SM3–SM12
B_{mapped}	Genes in (B) present in the expression matrix	SM3
N_B	Number of mapped genes within BP (B)	SM3, SM6, SM10
K	Number of selected BP observables used for patient-level scoring	SM7, SM12
$O_{B,p}$	Patient-level BP observable cloud $\{Z_{g,p}: g \in B\}$	SM5
$\mu_{B,p}$	Patient-level BP activity score	SM4
$W_{B,p}$	Patient-level within-BP observable width	SM5

Symbol	Meaning	Used in
$\bar{W}_B$	Cohort-level average of $W_{B,p}$	SM6
$D_{W-PHY,p}$	Patient-level stratification score	SM7
$D_{clinical}$	Survival-based discriminability score	SM8



Supplementary Figure S1. Overall analytical workflow of the D-PHY-I analysis.

This figure summarizes the complete analytical workflow used in the D-PHY-I study. For each cancer cohort, bulk transcriptomic expression data, matched clinical survival data, and structured pathway gene sets (Hallmark, GO:BP, and Reactome) were first collected and preprocessed through sample matching, primary-tumor filtering, gene cleaning, z-score normalization, and quality-control assessment. Mapped pathway gene sets were then used to construct patient-level biological-process (BP) observables. For each pathway B and patient p, patient-level BP activity was quantified as $\mu_{B,p}$, patient-level observable width was quantified as $W_{B,p} = \mathrm{FWHM}(O_{B,p})$, and observable coherence was defined from the activity–width relationship. Importantly, $W_{B,p}$ represents the within-BP gene-expression width for patient p, rather than the FWHM of BP activity scores across patients. These patient-level quantities were further summarized into cohort-level activity–width geometry.

Uncertainty-aware discriminability analysis was then performed by constructing a patient-level score that combines BP activity magnitude and observable-width burden, followed by Kaplan–Meier survival analysis and log-rank testing to obtain the clinical discriminability score D_{clinical}. Two complementary sensitivity analyses were included: a λ-scan to evaluate uncertainty-penalty effects and a K-scale observation-lens analysis to assess how discriminability depends on the number of selected BP observables. Validation analyses were further carried out using size-matched random gene-set baselines, random-background thresholding, D + coherence filtering, and biological-overlap analysis of high-discriminability random sets using Jaccard overlap. Finally, cross-cancer summaries, figure outputs, and reproducibility resources were integrated to support interpretation of process-level transcriptomic observables as uncertainty-bearing projected measurements in partially observable cancer systems.

SM1 | Data sources, cancer cohorts, and survival endpoints

This study analyzed bulk tumor transcriptomic profiles and matched clinical survival annotations from The Cancer Genome Atlas (TCGA). TCGA gene-expression and clinical survival data were obtained through the UCSC Xena platform. Each cancer cohort was processed independently to preserve cohort-specific transcriptomic structure and to avoid artificial effects introduced by pooled cross-cancer normalization.

The analysis was designed to evaluate process-level gene-expression observables under partial observability. Therefore, the primary input for each cancer cohort consisted of two matched data objects:

(i) a gene-expression matrix, with genes represented as rows and patient samples represented as columns, and

(ii) a clinical survival table containing patient identifiers, survival time, and event-status information. Patient matching between molecular and clinical records was performed using standardized TCGA patient barcodes.

The following TCGA cancer cohorts were included in the cross-cancer analysis: BLCA, BRCA, COAD, GBM, HNSC, KIRC, LAML, LIHC, LUAD, LUSC, OV, PAAD, PRAD, SARC, SKCM, STAD, THCA, and UCEC.

These cohorts were selected to provide a heterogeneous cross-cancer evaluation of process-level observable geometry and uncertainty-aware discriminability. The main manuscript uses these cohorts to assess cancer-specific responses to uncertainty weighting, cross-cancer activity–width organization, structured-versus-random behavior, and observation-scale effects. The main text specifically reports that the framework was applied across multiple cancer gene-expression cohorts and that cancer-specific λ responses, activity–width landscapes, random-baseline diagnostics, and K-scale effects were evaluated in Figures 2–5.

For each cohort, TCGA sample identifiers were standardized by retaining the first 12 characters of the barcode, corresponding to the patient-level identifier. When sample-type information was available, primary tumor samples were retained for analysis. Samples without valid survival-time or event-status information were excluded from survival-based discriminability analyses. After transcriptomic and clinical preprocessing, only patients present in both the gene-expression matrix and the survival table were retained.

Clinical survival endpoints were used as operational downstream outcome measures for evaluating discriminability. Survival time and event status were parsed from the available TCGA clinical survival files. When multiple survival-related fields were available, overall survival was prioritized for the main analysis. Event status was harmonized into binary form, with 0 representing censored observations and 1 representing observed events. Cohorts or patient groups with insufficient survival information were excluded from survival-based analyses.

Survival-based discriminability was evaluated using Kaplan–Meier survival analysis and the log-rank test after stratifying patients according to patient-level observable scores. The resulting log-rank *p*-value was transformed into the clinical discriminability score:

$$D_{\mathrm{clinical}} = -\log_{10}(p_{\mathrm{logrank}}).$$

This definition follows the main manuscript, where D_{clinical} is used to quantify the outcome-separation ability of uncertainty-aware patient-level scores. Importantly, survival was treated as an operational clinical endpoint rather than a direct measurement of latent biological-state evolution. The main manuscript also emphasizes that $D_{W\text{-}PHY,p}$ is a patient-level stratification score, whereas D_{clinical} is the survival-based discriminability value obtained from log-rank testing.

Several quality-control rules were applied to reduce unstable survival estimates. Samples with missing survival time, missing event status, duplicated patient identifiers, or non-positive survival time were removed. Survival analysis was not performed when the number of matched patients, event counts, or median-split group sizes were insufficient for stable Kaplan–Meier and log-rank evaluation. These constraints were used to reduce artifacts caused by sparse survival structure, especially in cancer cohorts with limited event numbers.

All cancer cohorts were analyzed independently throughout preprocessing, gene-set mapping, observable construction, uncertainty-width estimation, survival analysis, random-baseline generation, and figure construction. No expression values were pooled across cancer types before normalization or downstream analysis. This cohort-wise design ensured that cross-cancer comparisons reflected differences in process-level observable geometry and discriminability rather than artifacts of pooled normalization.

In summary, SM1 defines the data foundation of the study: independently processed TCGA cancer cohorts, matched gene-expression and survival data, patient-level barcode harmonization, survival endpoint parsing, and cohort-level quality-control criteria. The downstream construction of biological-process activity scores, patient-level observable widths, uncertainty-aware scores, and structured-versus-random validation procedures is described in SM2–SM13.

SM2 | Gene-expression preprocessing and cohort-wise normalization

For each TCGA cancer cohort, gene-expression matrices were preprocessed using a standardized cohort-wise workflow before biological-process observable construction. The objective of preprocessing was to generate a matched, gene-level, z-score-normalized transcriptomic matrix suitable for downstream biological-process activity scoring, patient-level observable-width estimation, uncertainty-aware discriminability analysis, and random-baseline validation.

SM2.1 Expression matrix loading

For each cancer cohort, the transcriptomic expression matrix was loaded as a gene-by-sample matrix. Rows represented genes, and columns represented TCGA tumor samples. Expression files were detected from cohort-specific data folders using standardized file names when available, including:

GE.tsv,HiSeqV2.tsv,expression.tsv.

The raw expression matrix for each cohort was represented as:

$$X \in \mathbb{R}^{G \times N},$$

where G denotes the number of genes and N denotes the number of patient samples before clinical matching and filtering.

All cancer cohorts were processed independently. No expression values were pooled across cancer types before normalization. This design was used to preserve cancer-specific transcriptomic structure and to prevent cross-cancer normalization artifacts. This is consistent with the main manuscript, where cancer-specific observable geometry and λ-dependent discriminability responses were evaluated across cohorts rather than after pooled normalization.

SM2.2 Gene-symbol standardization

Gene identifiers were standardized before downstream analysis. When gene identifiers contained annotation suffixes, transcript identifiers, or delimiter-based extensions, only the gene-symbol component was retained. Gene symbols were trimmed and converted to a consistent uppercase format.

For example:

TP53|7157 → TP53,

egfr → EGFR.

After standardization, duplicated gene symbols were collapsed by mean aggregation. If a gene g appeared m times in the expression matrix, the merged expression value was computed as:

$$X_{g,p} = \frac{1}{m} \sum_{i=1}^{m} X_{g_i,p},$$

where $X_{g_i,p}$ denotes the expression value of duplicated entry i for gene g in patient p.

This produced a gene-expression matrix with unique standardized gene symbols.

SM2.3 TCGA barcode harmonization and patient matching

TCGA sample identifiers were standardized using the first 12 characters of the TCGA barcode:

TCGA-XX-YYYY-01A... → TCGA-XX-YYYY.

This patient-level identifier was used to match transcriptomic samples with clinical survival annotations. When duplicated patient identifiers were present after barcode truncation, duplicated entries were removed or collapsed to preserve one molecular profile per patient.

Let P_{expr} denote the set of patient identifiers present in the expression matrix, and let P_{clin} denote the set of patient identifiers present in the clinical survival table. The matched patient set was defined as:

$$P_{\text{matched}} = P_{\text{expr}} \cap P_{\text{clin}}.$$

Only patients present in both expression and survival data were retained for survival-based discriminability analysis.

SM2.4 Primary tumor sample filtering

When sample-type information was available from TCGA barcodes or phenotype annotations, primary tumor samples were retained for analysis. TCGA primary tumor samples are typically indicated by sample type code “01” in the barcode.

Samples corresponding to non-primary tumor tissues, metastatic samples, recurrent tumors, adjacent normal tissues, or unmatched sample types were excluded when identifiable. This filtering step was used to reduce heterogeneity caused by mixing distinct tissue or sampling categories.

After filtering, each cancer cohort contained a matched primary tumor expression matrix and a corresponding survival annotation table.

SM2.5 Clinical survival-table matching

Clinical survival data were matched to the transcriptomic matrix after barcode harmonization. Survival tables were parsed to identify survival time and event-status columns. Overall survival was prioritized as the primary endpoint when available.

Survival time was required to be positive and non-missing. Event status was harmonized into binary form:

$$0 = \text{censored},$$
$$1 = \text{event}.$$

Samples with missing survival time, missing event status, duplicated patient identifiers, or non-positive survival time were excluded. This ensured that all retained patients had both transcriptomic observations and valid survival annotations for downstream Kaplan–Meier and log-rank testing.

SM2.6 Cohort-wise gene-level z-score normalization

After sample matching and filtering, expression values were normalized independently within each cancer cohort. For each gene g, expression values were *z*-score normalized across matched patients:

$$Z_{g,p} = \frac{X_{g,p} - \bar{X}_g}{s_g},$$

where $X_{g,p}$ is the expression value of gene g in patient p, $\bar{X}_g$ is the mean expression of gene g across patients within the same cancer cohort, and s_g is the standard deviation of gene g across patients within that cohort.

The resulting normalized expression matrix was:

$$Z \in \mathbb{R}^{G \times N},$$

where G is the number of standardized genes and N is the number of matched patients retained for that cancer cohort.

This within-cohort normalization ensured that downstream biological-process scores represented relative transcriptomic activity within each cancer cohort rather than absolute expression-scale differences between genes or cancer types. The main manuscript uses these z-score-normalized gene-expression values to define patient-level BP observables $O_{B,p}$, activity scores $\mu_{B,p}$, and observable widths $W_{B,p}$.

SM2.7 Handling of zero-variance genes and missing values

Genes with zero variance across patients cannot be meaningfully z-score normalized. To avoid numerical instability, zero-variance genes and invalid normalized values were handled using stable replacement procedures. Infinite values and undefined values generated during normalization were converted to missing values and then replaced with zero where appropriate.

This step prevented unstable downstream averaging, FWHM estimation, and random gene-set construction.

SM2.8 Quality-control thresholds

Several quality-control thresholds were applied before downstream analysis. A cohort or survival-based comparison was excluded when the number of matched patients, number of observed events, or size of survival-stratified groups was insufficient for stable analysis.

The following general criteria were applied:

$$N \geq 30,$$

$$E \geq 5,$$

where N is the number of matched patients and E is the number of observed survival events. For median-split survival analysis, each stratified group was required to contain a sufficient number of patients, typically:

$$n_{\text{group}} \geq 10.$$

These criteria were used to reduce unstable Kaplan–Meier and log-rank estimates caused by sparse survival structure.

SM2.9 Output of preprocessing

The preprocessing stage produced the following output objects for each cancer cohort:

$$Z,$$

the cohort-wise z-score-normalized gene-expression matrix;

$$P_{\text{matched}},$$

the matched patient set;

$$S = \{(p_i, t_i, e_i)\}_{i=1}^{N},$$

the survival annotation table, where p_i denotes patient identifier, t_i denotes survival time, and e_i denotes binary event status; and the cohort-specific gene universe:

$$G_{\text{expr}} = \{g_1, g_2, \dots, g_G\}.$$

These outputs were used as direct inputs for biological-process gene-set mapping in SM3, patient-level BP activity construction in SM4, patient-level observable-width estimation in SM5, and size-matched random gene-set generation in SM10.

SM2.10 Summary

In summary, SM2 defines the preprocessing workflow used to transform raw TCGA gene-expression matrices into cohort-wise normalized transcriptomic observation matrices. Each cancer cohort was processed independently through gene-symbol standardization, barcode harmonization, primary tumor filtering, survival matching, quality control, and gene-level z-score normalization. The resulting normalized matrix Z provided the common transcriptomic basis for all downstream activity–width, discriminability, random-baseline, and observation-scale analyses.

SM3 | Biological-process and pathway gene-set mapping

Biological-process and pathway gene sets were used to construct process-level transcriptomic observables from the cohort-wise normalized gene-expression matrices described in SM2. The purpose of this step was to map high-dimensional gene-expression profiles onto biologically organized gene sets, allowing each process-level observable to be evaluated by activity, observable width, coherence, discriminability, and random-background behavior.

SM3.1 Gene-set collections

The primary structured gene-set collections used in this study were:

Hallmark,
GO Biological Process (GO:BP),
Reactome.

These collections were selected because they provide complementary levels of biological-process and pathway organization. Hallmark gene sets represent compact biological programs, GO:BP provides broader biological-process annotations, and Reactome provides pathway-oriented functional annotations. The main manuscript uses these structured biological-process observables to construct activity–width landscapes, compare cancer-by-database observable width, evaluate structured-versus-random behavior, and perform scale-dependent discriminability analysis.

Additional pathway collections may be prepared in the computational repository for extended analyses, but the primary AIDO-D-PHY-I analyses and main-text figures focus on Hallmark, GO:BP, and Reactome unless otherwise stated.

SM3.2 GMT file parsing

Gene sets were loaded from Gene Matrix Transposed format files. Each GMT entry was parsed as:

$$\text{GeneSetNameDescription} g_1 g_2 \cdots g_m.$$

For each gene-set entry, the pathway or biological-process name was extracted together with its listed gene members. Gene symbols were standardized using the same gene-symbol normalization procedure described in SM2, including trimming of whitespace, removal of suffixes where applicable, and conversion to uppercase symbols.

Each gene-set collection was represented as:

$$\mathcal{B} = \{B_1, B_2, \dots, B_M\},$$

where B_i denotes one biological-process or pathway gene set, and M denotes the number of gene sets in the collection.

SM3.3 Mapping gene sets to the cohort-specific expression universe

For each cancer cohort, gene sets were mapped to the available cohort-specific expression gene universe:

$$G_{\text{expr}} = \{g_1, g_2, \dots, g_G\},$$

where G_{expr} was obtained from the z-score-normalized expression matrix Z after preprocessing.

For each biological-process gene set B, the mapped gene set was defined as:

$$B_{\text{mapped}} = B \cap G_{\text{expr}}.$$

Only genes present in both the gene-set definition and the processed expression matrix were retained for downstream analysis.

SM3.4 Definition of mapped gene-set size N_B

For each mapped biological-process observable B, the mapped gene-set size was defined as:

$$N_B = | B_{\text{mapped}} |.$$

Here, N_B denotes the number of mapped genes inside a single BP observable. It describes the gene-set aggregation scope used to construct the patient-level BP observable:

$$O_{B,p} = \{Z_{g,p} : g \in B_{\text{mapped}}\}.$$

This quantity is distinct from K, which is introduced later in SM12. N_B refers to the number of genes within one biological-process observable, whereas K refers to the number of selected BP observables used to construct a patient-level uncertainty-aware score. This distinction is important because Figure 3D evaluates how observable width depends on gene-set aggregation scope N_B, whereas Figure 5 evaluates how discriminability depends on the number of selected BP observables K. The main manuscript explicitly separates these two scale levels when interpreting Figure 3D and Figure 5.

SM3.5 Minimum mapped-gene threshold

Gene sets with too few mapped genes were excluded because very small gene sets provide unstable estimates of process-level activity and FWHM-defined observable width. The minimum mapped-gene threshold was:

$$N_B \geq 5.$$

Gene sets failing this threshold were removed from downstream activity–width and discriminability analyses.

This threshold ensured that each retained biological-process observable contained enough genes to estimate both the patient-level activity score $\mu_{B,p}$ and patient-level observable width $W_{B,p}$.

SM3.6 Controlled mapped-gene-size range

For analyses requiring stricter control of gene-set size effects, including structured-versus-random comparison and some width-scale analyses, a controlled mapped-gene-size range was used:

$$10 \leq N_B \leq 300.$$

This controlled range was introduced to reduce two opposite sources of instability. Very small gene sets may produce unstable activity and width estimates because they contain too few mapped genes. Very large gene sets may dilute coherent process-level signals by averaging across broad or heterogeneous biological programs.

The controlled range was especially important for random-baseline analysis, where size-matched random gene sets were generated using the same N_B as the corresponding structured biological-process gene set. The random-baseline workflow is described in SM10.

SM3.7 Construction of mapped BP observable list

After gene-set mapping and filtering, each cancer cohort and gene-set collection produced a retained list of mapped biological-process observables:

$$\mathcal{B}_{\text{retained}} = \{B_1, B_2, \dots, B_{M'}\},$$

where M' denotes the number of retained mapped gene sets after filtering.

Each retained BP observable contained:

$$B_{\text{mapped}},$$

$$N_B,$$

and the corresponding expression submatrix:

$$Z_B = \{Z_{g,p}: g \in B_{\text{mapped}}, p \in P_{\text{matched}}\}.$$

This mapped expression submatrix served as the direct input for the patient-level BP activity score in SM4 and the patient-level observable width in SM5.

SM3.8 Relationship between gene-set mapping and observable construction

The mapping step does not itself define biological-process activity or uncertainty width. Instead, it defines the gene membership of each process-level observable. For each retained BP gene set B, the mapped genes determine the patient-level observable cloud:

$$O_{B,p} = \{Z_{g,p}: g \in B_{\text{mapped}}\}.$$

The center of this cloud is quantified by $\mu_{B,p}$, defined in SM4, and its width is quantified by $W_{B,p}$, defined in SM5. Thus, gene-set mapping provides the biological and computational boundary within which patient-level process activity and observable width are calculated.

This is consistent with the main manuscript, where each BP observable is treated as a collective process-level measurement derived from grouped gene-expression signals rather than as a single-gene or scalar-only feature.

SM3.9 Output of gene-set mapping

The gene-set mapping stage generated the following output objects for each cancer cohort and gene-set collection:

$$\mathcal{B}_{\text{retained}},$$

the retained list of mapped biological-process observables;

$$B_{\text{mapped}},$$

the mapped gene members for each BP observable;

$$N_B,$$

the mapped gene-set size for each BP observable; and

$$Z_B,$$

the expression submatrix associated with each mapped BP observable.

These outputs were used for patient-level BP activity calculation in SM4, patient-level observable-width estimation in SM5, cohort-level activity–width summaries in SM6, size-matched random baseline construction in SM10, and gene-set aggregation scope analysis in Figure 3D.

SM3.10 Summary

In summary, SM3 defines how structured biological-process and pathway gene sets were mapped onto cohort-specific TCGA expression matrices. Hallmark, GO:BP, and Reactome gene-set collections were parsed from GMT files, standardized, intersected with the cohort-specific gene universe, and filtered by mapped gene-set size. The mapped gene-set size N_B was defined as the number of retained genes within each BP observable. This quantity determines the gene-set aggregation scope of each process-level observable and should be distinguished from K, the number of BP observables selected for patient-level score construction in the observation-lens analysis.

SM4 | Patient-level BP activity score, $\boldsymbol{\mu_{B,p}}$

After biological-process and pathway gene sets were mapped to the cohort-specific gene-expression universe, patient-level BP activity scores were computed from the z-score-normalized expression matrix Z. The purpose of this step was to summarize the relative transcriptomic activity of each biological-process observable in each patient while preserving a clear distinction between activity center and observable width.

SM4.1 Input from gene-set mapping

For each cancer cohort, the preprocessing workflow described in SM2 generated a cohort-wise normalized expression matrix:

$$Z \in \mathbb{R}^{G \times N},$$

where G denotes the number of genes and Ndenotes the number of matched patients.

From SM3, each retained biological-process gene set B was represented by its mapped gene set:

$$B_{\text{mapped}} = B \cap G_{\text{expr}},$$

with mapped gene-set size:

$$N_B = | B_{\text{mapped}} |.$$

For patient p, the mapped expression values associated with biological process Bwere extracted as:

$$\{Z_{g,p} : g \in B_{\text{mapped}}\}.$$

This mapped gene-expression set provided the basis for computing the patient-level BP activity score.

SM4.2 Definition of patient-level BP activity score

For each biological process B and patient p, the patient-level BP activity score was defined as the mean z-score-normalized expression of genes mapped to B:

$$\mu_{B,p} = \frac{1}{N_B} \sum_{g \in B_{\text{mapped}}} Z_{g,p} \, .$$

Here, $Z_{g,p}$denotes the z-score-normalized expression value of gene g in patient p, and N_B denotes the number of mapped genes in biological process B.

Thus, $\mu_{B,p}$ summarizes the center of the mapped process-level gene-expression signal for one patient. This definition matches the main manuscript, where patient-level BP activity is computed as the average z-score-normalized expression of genes mapped to process B.

SM4.3 Interpretation of $\boldsymbol{\mu_{B,p}}$

Because $\mu_{B,p}$ is computed from cohort-wise z-score-normalized expression values, it should be interpreted as a signed relative activity score rather than an absolute biological activity measurement.

A positive value of $\mu_{B,p}$ indicates that, on average, genes mapped to biological process B show higher-than-cohort-average expression in patient p. A negative value indicates lower-than-cohort-average expression. A value near zero indicates that the mapped genes are close to their cohort-level mean expression.

Therefore, $\mu_{B,p}$ represents relative process-level transcriptomic activity within a cancer cohort. It should not be interpreted as a direct measurement of pathway activation, enzyme activity, protein signaling strength, or latent biological-state value.

SM4.4 Activity center of the patient-level BP observable

The BP activity score $\mu_{B,p}$ was interpreted as the activity center of the patient-level BP observable. For a given biological process B, the mapped gene-expression values in patient $p$define a process-level observation set. The mean of this set provides the activity center:

$$\mu_{B,p} = \mathrm{center}(O_{B,p}).$$

The full definition of $O_{B,p}$ and its FWHM-defined width $W_{B,p}$ is provided in SM5. In the present section, $\mu_{B,p}$ is defined only as the activity-center component of the patient-level BP observable.

This separation is important because the proposed activity–width framework evaluates process-level observables using two complementary quantities: activity center, $\mu_{B,p}$, and observable width, $W_{B,p}$. The main manuscript explicitly distinguishes these quantities when defining patient-level BP observables and cohort-level activity–width summaries.

SM4.5 Construction of the BP activity matrix

For each cancer cohort and gene-set collection, patient-level BP activity scores were computed for all retained biological-process observables. This produced a BP activity matrix:

$$\mathbf{M}_{\mu} = [\mu_{B,p}] \in \mathbb{R}^{M\times N},$$

where M denotes the number of retained BP observables and N denotes the number of matched patients.

Each row of $\mathbf{M}_{\mu}$ corresponds to one biological-process observable, and each column corresponds to one patient.

This matrix was used in downstream analyses to compute cohort-level activity summaries, uncertainty-aware patient-level scores, survival-based discriminability, and K-scale observation-lens analysis.

SM4.6 Absolute activity magnitude

Because $\mu_{B,p}$ is signed, both positive and negative values may represent deviations from the cohort-level reference. For analyses focused on activity magnitude rather than direction, the absolute value was used:

$$|\ \mu_{B,p}\ |.$$

This quantity measures the magnitude of relative process-level deviation regardless of whether the mapped BP genes are expressed above or below the cohort mean.

The absolute activity magnitude was used in the patient-level magnitude term:

$$M_p = \frac{1}{K}\sum_{B=1}^{K} |\mu_{B,p}|,$$

which is formally defined later in SM7. The main manuscript uses the same logic when defining M_p as the average activity-magnitude term across selected BP observables.

SM4.7 Cohort-level activity summary

For cohort-level visualization and comparison across BP observables, patient-level activity scores were summarized across all matched patients.

The cohort-level average activity magnitude of biological process $B$was defined as:

$$|\bar{\mu}|_B = \frac{1}{N}\sum_{p=1}^{N} |\mu_{B,p}|.$$

Here, N denotes the number of matched patients in the cancer cohort.

This quantity was used to position BP observables along the activity axis in cohort-level activity–width landscapes. In the main manuscript, each BP gene set is represented in cohort-level $|\bar{\mu}|_B$–$\bar{W}_B$ space to compare average relative activity magnitude and average observable width across biological processes.

SM4.8 Relationship between $\boldsymbol{\mu_{B,p}}$, $|\bar{\boldsymbol{\mu}}|_{\boldsymbol{B}}$, and downstream analysis

The patient-level score $\mu_{B,p}$ and the cohort-level summary $|\bar{\mu}|_B$ play different roles.

$$\mu_{B,p}$$

is a patient-level BP activity score used for patient-level scoring, survival stratification, and uncertainty-aware discriminability analysis.

$$|\bar{\mu}|_B$$

is a cohort-level summary used for visualization and comparison of BP observables in activity–width space.

Thus, $\mu_{B,p}$ should not be confused with the cohort-level activity summary. The former is patient-specific, whereas the latter summarizes the average activity magnitude of a BP observable across the cohort.

SM4.9 Output of patient-level BP activity scoring

The BP activity-scoring step generated the following output objects for each cancer cohort and gene-set collection:

$$\mathbf{M}_\mu = [\mu_{B,p}],$$

the patient-level BP activity matrix;

$$|\,\mathbf{M}_\mu\,| = [|\,\mu_{B,p}\,|],$$

the absolute activity-magnitude matrix; and

$$|\,\bar{\mu}\,|_B,$$

the cohort-level average activity magnitude for each BP observable.

These outputs were used in SM6 to construct cohort-level activity–width summaries and coherence measures, in SM7 to construct uncertainty-aware patient-level scores, and in SM12 to perform K-scale observation-lens analysis.

SM4.10 Summary

In summary, SM4 defines the patient-level BP activity score $\mu_{B,p}$ as the mean z-score-normalized expression of mapped genes within biological process B for patient p. Because the input expression values are cohort-wise normalized, $\mu_{B,p}$ is interpreted as a signed relative process-level activity score. Its absolute value, $|\,\mu_{B,p}\,|$, was used when activity magnitude rather than direction was required. Cohort-level activity magnitude was summarized as $|\,\bar{\mu}\,|_B$ for activity–width visualization and coherence analysis. The corresponding patient-level observable width $W_{B,p}$ is defined separately in SM5.

SM5 | Patient-level observable width, $W_{B,p}$

After computing the patient-level BP activity score $\mu_{B,p}$, the observable width of each biological-process gene set was estimated at the patient level. The purpose of this step was to quantify the distributional width of the mapped gene-expression values inside each biological process for each patient.

This section defines the patient-level BP observable cloud $O_{B,p}$, the FWHM-defined observable width $W_{B,p}$, and the operational interpretation of $W_{B,p}$ as process-level observable cloud thickness.

SM5.1 Patient-level BP observable cloud

For each biological process B and patient p, the mapped genes of B define a patient-level BP observable cloud:

$$O_{B,p} = \{Z_{g,p}: g \in B_{\mathrm{mapped}}\}.$$

Here, $Z_{g,p}$ denotes the z-score-normalized expression value of gene g in patient p, and B_{mapped} denotes the set of genes from biological process B that are present in the cohort-specific expression matrix.

Thus, $O_{B,p}$ is the set of *z*-score-normalized expression values of all mapped genes belonging to biological process B in one patient.

This definition follows the main manuscript, where each patient-level BP observable is represented as:

$$O_{B,p} = \{Z_{g,p}: g \in B\}.$$

The corresponding BP activity score $\mu_{B,p}$ is the center of this observable cloud, while $W_{B,p}$ quantifies its distributional width.

SM5.2 Definition of patient-level observable width

For each patient-level BP observable cloud $O_{B,p}$, the observable width was estimated using full width at half maximum:

$$W_{B,p} = \mathrm{FWHM}(O_{B,p}).$$

Therefore, $W_{B,p}$ measures the width of the gene-expression distribution formed by the mapped genes of biological process B in patient p.

This definition is central to the revised D-PHY-I framework. In the present manuscript, $W_{B,p}$ is not defined as the FWHM of BP activity scores across patients. Instead, it is defined as the FWHM of the within-process gene-expression distribution for a single patient.

In other words, the following interpretation was not used in the revised manuscript:

$$W_B \neq \mathrm{FWHM}\big(P(\mu_{B,p})\big).$$

Instead, the analysis used:

$$W_{B,p} = \mathrm{FWHM}\big(\{Z_{g,p}: g \in B_{\mathrm{mapped}}\}\big).$$

This distinction is important because the main text defines W as patient-level BP observable width, while cohort-level summaries are obtained only after $W_{B,p}$ has been computed for each patient.

SM5.3 Relationship between $\boldsymbol{\mu_{B,p}}$ and $\boldsymbol{W_{B,p}}$

The patient-level BP observable $O_{B,p}$ is summarized by two complementary quantities:

$$O_{B,p} \Rightarrow (\mu_{B,p}, W_{B,p}).$$

The activity score $\mu_{B,p}$, defined in SM4, represents the center of the mapped gene-expression distribution:

$$\mu_{B,p} = \frac{1}{N_B} \sum_{g \in B_{\text{mapped}}} Z_{g,p} \,.$$

The observable width $W_{B,p}$ represents the FWHM-defined width of the same mapped gene-expression distribution:

$$W_{B,p} = \text{FWHM}(O_{B,p}).$$

Thus, $\mu_{B,p}$ and $W_{B,p}$ are calculated from the same patient-level gene-expression set but describe different aspects of the observable. $\mu_{B,p}$ describes relative activity center, whereas $W_{B,p}$ describes distributional spread.

This separation allows two BP observables with similar activity scores to be distinguished if they differ in observable width, compactness, or coherence.

SM5.4 FWHM estimation procedure

For each $O_{B,p}$, the mapped gene-expression values were treated as an empirical distribution. The FWHM was estimated from this distribution using a density-based procedure when possible.

The procedure consisted of the following steps:

1. Extract the mapped gene-expression values:

$$O_{B,p} = \{Z_{g,p}: g \in B_{\text{mapped}}\}.$$

2. Estimate the empirical or kernel-smoothed density of $O_{B,p}$.
3. Identify the maximum density value:

$$f_{\max} = \max_{x} \; f_{B,p}(x).$$

4. Define the half-maximum level:

$$f_{\text{half}} = \frac{1}{2} f_{\max}.$$

5. Identify the left and right boundaries at which the density crosses the half-

maximum level:

$$x_{\text{left}}, x_{\text{right}}.$$

6. Compute the FWHM-defined observable width:

$$W_{B,p} = x_{\text{right}} - x_{\text{left}}.$$

When density estimation was unstable because of small mapped gene-set size, repeated values, or numerical singularity, a robust fallback width estimator was used to prevent missing or unstable width estimates.

SM5.5 Fallback width estimation

For gene sets where KDE-based FWHM estimation failed or was numerically unstable, an interquartile-range-based fallback approximation was used.

The interquartile range was defined as:

$$IQR_{B,p} = Q_{75}(O_{B,p}) - Q_{25}(O_{B,p}).$$

The fallback width was approximated as:

$$W_{B,p} \approx 2.355 \times \frac{IQR_{B,p}}{1.349}.$$

This approximation maps the interquartile range to an approximate Gaussian-equivalent FWHM scale. The fallback was used only to maintain numerical stability and does not change the operational interpretation of $W_{B,p}$ as a width estimate of the patient-level BP observable cloud.

SM5.6 Numerical stabilization

To avoid instability in downstream logarithmic transformations and coherence calculations, a small positive stabilization constant was used:

$$\delta = 10^{-6}.$$

This constant was applied in later calculations involving $W_{B,p}$, including:

$$\log{(W_{B,p} + \delta)},$$

and

$$\frac{\mid \mu_{B,p} \mid}{W_{B,p} + \delta}.$$

The stabilization constant was not used to redefine $W_{B,p}$; it was only used to prevent division-by-zero and logarithmic instability in downstream analyses.

SM5.7 Operational interpretation of $W_{B,p}$

Within the present framework, $W_{B,p}$ was interpreted as an operational measure of observable width or cloud thickness for biological process B in patient p.

A small $W_{B,p}$ indicates that mapped genes within biological process B form a relatively compact expression distribution in patient p. A large $W_{B,p}$ indicates that the mapped genes form a broader or more diffuse expression distribution.

Importantly, $W_{B,p}$ was not interpreted as literal thermodynamic entropy, direct physical disorder, or a direct measurement of hidden biological complexity. Instead, it was treated as a reproducible proxy for the dispersion of the projected process-level transcriptomic signal. This interpretation is consistent with the main manuscript, which states that W is an operational measure of observable cloud thickness rather than literal entropy.

Potential contributors to broader $W_{B,p}$ include:

gene-set coordination looseness,
heterogeneous transcriptional regulation,
cellular mixture effects,
projection ambiguity,
biological fluctuation,
measurement variability.

The present analysis did not attempt to decompose these sources. Instead, $W_{B,p}$ was used as an operational width descriptor of the observed BP-level gene-expression cloud.

SM5.8 Width matrix construction

For each cancer cohort and gene-set collection, $W_{B,p}$ was computed for all retained BP observables and all matched patients.

This produced a patient-level observable-width matrix:

$$\mathrm{M}_W = [W_{B,p}] \in \mathbb{R}^{M \times N},$$

where M denotes the number of retained BP observables and N denotes the number of matched patients.

Each row corresponds to one biological-process observable, and each column corresponds to one patient.

This matrix was used in later analyses to compute cohort-level width summaries, coherence values, uncertainty burden U_p, uncertainty-aware patient-level scores, and K-scale observation-lens analyses.

SM5.9 Cohort-level width summary

Although $W_{B,p}$ was defined at the patient level, cohort-level visualization required a summary width for each BP observable.

For biological process B, the cohort-level average observable width was defined as:

$$\bar{W}_B = \frac{1}{N}\sum_{p=1}^{N} W_{B,p}\,.$$

Here, N denotes the number of matched patients in the cancer cohort.

This quantity was used to position each BP observable in cohort-level activity–width space:

$$B \Rightarrow (|\bar{\mu}|_B, \bar{W}_B).$$

The cohort-level summary $\bar{W}_B$ should therefore be understood as the average of patient-level within-BP widths, not as the FWHM of patient-level BP activity scores across the cohort.

This distinction directly supports the main manuscript's revised Figure 3 logic, where BP observables are compared using cohort-level activity–width summaries derived from patient-level $\mu_{B,p}$ and $W_{B,p}$.

SM5.10 Difference from across-patient activity dispersion

To avoid ambiguity, the present analysis distinguishes two different types of dispersion:

Patient-level within-BP width

$$W_{B,p} = \mathrm{FWHM}\big(\{Z_{g,p}: g \in B_{\mathrm{mapped}}\}\big).$$

This is the width used in the present manuscript.

Across-patient BP activity dispersion

$$\mathrm{FWHM}\big(\{\mu_{B,p}: p = 1, \dots, N\}\big).$$

This quantity describes how BP activity scores vary across patients, but it was not used as the definition of $W_{B,p}$ in the revised D-PHY-I framework.

The first quantity describes the internal gene-expression width of one BP observable in one patient. The second quantity describes cohort-level variation of scalar BP activity scores. These are conceptually and mathematically different.

The main manuscript uses the first definition because the goal is to represent each patient-level BP observable as an activity–width pair:

$$O_{B,p} \Rightarrow (\mu_{B,p}, W_{B,p}).$$

SM5.11 Output of patient-level observable-width estimation

The observable-width estimation step generated the following outputs for each cancer cohort and gene-set collection:

$$\mathrm{M}_W = [W_{B,p}],$$

the patient-level BP observable-width matrix;

$$\bar{W}_B,$$

the cohort-level average observable width for each BP observable; and width-related numerical summaries, including median width, width variability, and quality-control indicators when required.

These outputs were used in SM6 to construct cohort-level activity–width summaries and coherence measures, in SM7 to compute uncertainty burden U_p, in SM10–SM11 to evaluate structured-versus-random and D + coherence filtering, and in SM12 to evaluate observation-lens effects.

SM5.12 Summary

In summary, SM5 defines $W_{B,p}$ as the FWHM-defined width of the patient-level BP observable cloud:

$$O_{B,p} = \{Z_{g,p}: g \in B_{\text{mapped}}\},$$
$$W_{B,p} = \mathrm{FWHM}(O_{B,p}).$$

This definition means that $W_{B,p}$ measures the within-process gene-expression distributional width for biological process B in patient p. Cohort-level width summaries were computed only after patient-level widths had been estimated:

$$\bar{W}_B = \frac{1}{N}\sum_{p=1}^{N} W_{B,p}\,.$$

Thus, $W_{B,p}$ is a patient-level observable-width measure, while $\bar{W}_B$ is a cohort-level summary used for visualization and comparison. This distinction is essential for aligning the Supplementary Methods with the revised main text.

SM6 | Cohort-level activity–width summaries and coherence

After patient-level BP activity scores $\mu_{B,p}$ and patient-level observable widths $W_{B,p}$ were computed, cohort-level summaries were constructed for each biological-process observable. These summaries were used to compare BP observables across cancer cohorts, pathway databases, and mapped gene-set sizes.

The purpose of this section is to define the cohort-level activity–width representation, coherence score, and gene-set-size-dependent summary statistics used in the main manuscript.

SM6.1 Input patient-level quantities

For each retained biological process B and patient p, two patient-level quantities were available from SM4 and SM5:

$$\mu_{B,p} = \frac{1}{N_B}\sum_{g\in B_{\text{mapped}}} Z_{g,p},$$

and

$$W_{B,p} = \mathrm{FWHM}\big(\{Z_{g,p}: g \in B_{\text{mapped}}\}\big).$$

Here, $\mu_{B,p}$ represents the patient-level BP activity center, and $W_{B,p}$ represents the patient-level within-BP observable width.

These quantities were computed for each retained BP observable across all matched patients in a cancer cohort.

SM6.2 Cohort-level average activity magnitude

Because $\mu_{B,p}$ is a signed relative activity score, the cohort-level activity magnitude of BP observable B was summarized using the average absolute activity score:

$$|\bar{\mu}|_B = \frac{1}{N}\sum_{p=1}^{N} |\mu_{B,p}|.$$

Here, N denotes the number of matched patients in the cancer cohort.

The use of $|\mu_{B,p}|$ reflects the magnitude of process-level deviation from the cohort reference, regardless of whether the mapped BP genes are expressed above or below the cohort mean. Thus, $|\bar{\mu}|_B$ represents the average relative activity magnitude of process B across patients.

This quantity was used as the activity axis in cohort-level activity–width landscapes.

SM6.3 Cohort-level average observable width

For each BP observable B, the cohort-level average width was computed as the mean of patient-level observable widths:

$$\bar{W}_B = \frac{1}{N}\sum_{p=1}^{N} W_{B,p}.$$

Importantly, $\bar{W}_B$ is the average of patient-level within-BP widths. It is not the

FWHM of BP activity scores across patients.

Thus, the following definition was used:

$$\bar{W}_B = \frac{1}{N}\sum_{p=1}^{N} \mathrm{FWHM}\left(\{Z_{g,p}: g \in B_{\mathrm{mapped}}\}\right),$$

whereas the following across-patient activity-distribution width was not used as the definition of $\bar{W}_B$:

$$\mathrm{FWHM}\left(\{\mu_{B,p}: p = 1, \ldots, N\}\right).$$

This distinction is essential because the main manuscript defines $W_{B,p}$ as patient-level BP observable width and then summarizes it across patients for cohort-level visualization.

SM6.4 Cohort-level activity–width representation

Each retained BP observable B was represented in cohort-level activity–width space as:

$$B \Rightarrow (|\ \bar{\mu}\ |_B, \bar{W}_B).$$

In this representation, $|\ \bar{\mu}\ |_B$ describes the average magnitude of the BP activity center across patients, whereas $\bar{W}_B$ describes the average observable cloud thickness of that BP across patients.

This representation was used to generate the cross-cancer activity–width landscape in Figure 3A of the main manuscript. Each point in this landscape represents one BP observable summarized by its cohort-level activity magnitude and average observable width. The main manuscript uses this representation to show that BP observables are not fully described by scalar activity alone, because observables with comparable activity magnitude may show different widths.

SM6.5 Patient-level coherence

For each BP observable B and patient p, patient-level coherence was defined as:

$$C_{B,p} = \frac{|\mu_{B,p}|}{W_{B,p} + \delta},$$

where $\delta = 10^{-6}$ is a small positive constant used for numerical stabilization.

This quantity measures the activity magnitude of a patient-level BP observable relative to its observable width. A larger value indicates stronger activity magnitude relative to width, whereas a smaller value indicates weaker activity magnitude, broader width, or both.

Coherence was interpreted as an operational activity–width compactness measure. It was not treated as a literal signal-to-noise ratio, because $W_{B,p}$ may reflect biological heterogeneity, gene-set coordination looseness, projection ambiguity, and measurement variability rather than technical noise alone.

SM6.6 Cohort-level coherence summary

For each BP observable B, cohort-level coherence was summarized as the mean patient-level coherence:

$$\bar{C}_B = \frac{1}{N}\sum_{p=1}^{N} C_{B,p} = \frac{1}{N}\sum_{p=1}^{N}\frac{|\mu_{B,p}|}{W_{B,p} + \delta}.$$

This coherence summary was used to compare BP observables according to the balance between activity magnitude and observable width.

A BP observable with high $\bar{C}_B$ tends to show relatively strong activity magnitude and compact width across patients. A BP observable with low $\bar{C}_B$ tends to show weak activity magnitude, broad width, or both.

SM6.7 Alternative cohort-level coherence for visualization

For some cohort-level visualizations, an aggregate coherence index was also computed from cohort-level summaries:

$$C_B^{\text{cohort}} = \frac{|\bar{\mu}|_B}{\bar{W}_B + \delta}.$$

This value provides a simple summary of the position of BP observable B in cohort-level activity–width space. It is useful for visualization and ranking but should be interpreted as a descriptive summary rather than a replacement for patient-level coherence.

When patient-level coherence and cohort-level coherence were both available, patient-level coherence was used for patient-level scoring and filtering, whereas cohort-level coherence was used for visualization, database-level summaries, and cross-cancer descriptive comparison.

SM6.8 Cancer-by-database width summaries

To compare observable-width behavior across cancer cohorts and pathway databases, $\bar{W}_B$ values were summarized within each cancer–database combination.

For cancer cohort $c$and gene-set database d, the database-level average width was computed as:

$$\bar{W}_{c,d} = \frac{1}{M_{c,d}} \sum_{B \in \mathcal{B}_{c,d}} \bar{W}_B,$$

where $\mathcal{B}_{c,d}$ denotes the retained BP observables for cancer $c$and database d, and $M_{c,d}$ denotes the number of retained BP observables in that combination.

This summary was used to generate Figure 3B. In the main manuscript, cancer-by-database width summaries were used to show that average observable width varies across cancer contexts and pathway databases. NA entries indicate cancer–database combinations without available summary entries in the integrated analysis output.

SM6.9 Coherence-based organization of cancer cohorts

To summarize cross-cancer differences in process-level observable compactness, coherence profiles were constructed for each cancer cohort.

For each cancer cohort, a vector of BP coherence summaries was generated:

$$\mathbf{C}_c = (\bar{C}_{B_1}, \bar{C}_{B_2}, \dots, \bar{C}_{B_m}),$$

where $B_1, \dots, B_m$ denote BP observables available for comparison.

These coherence profiles were used to organize cancer cohorts descriptively according to their process-level activity–width compactness. The resulting organization was used for Figure 3C.

This analysis was unsupervised and descriptive. It was not intended as supervised cancer classification, diagnostic labeling, or evidence of mechanistic equivalence between cancers. Instead, it was used to show that activity–width coherence profiles contain cross-cancer organizational information. This interpretation follows the main manuscript's description of Figure 3C.

SM6.10 Gene-set size dependence of observable width

To evaluate whether observable width depends on gene-set aggregation scope, each BP observable was grouped according to its mapped gene-set size:

$$N_B = | B_{\text{mapped}} |.$$

BP observables were assigned to gene-set-size bins. Within each size bin, the distribution of cohort-level widths $\bar{W}_B$ was summarized using the median or mean width.

For a gene-set-size bin s, the median width was computed as:

$$\widetilde{W}_s = \mathrm{median}(\bar{W}_B : N_B \in s).$$

This analysis was used to generate Figure 3D.

The purpose of this analysis was to test whether process-level observable width changes with the number of genes used to construct a BP observable. In the main manuscript, this was interpreted as gene-set aggregation scope: very small gene sets may incompletely capture collective process-level organization, whereas overly broad gene sets may dilute coherent signal structure by averaging across heterogeneous subprograms.

SM6.11 Distinction between N_B and K

Two different scale quantities were used in the study:

$$N_B = \text{number of mapped genes within biological process } B,$$

and

$$K = \text{number of selected BP observables used to construct a patient-level score.}$$

These two quantities describe different levels of scale.

N_B describes the gene-set aggregation scope of one BP observable. It is relevant to the construction of $O_{B,p}$, $\mu_{B,p}$, $W_{B,p}$, and the gene-set-size analysis shown in Figure 3D.

K describes the observable-set scale used when combining multiple BP observables to construct the patient-level uncertainty-aware score $D_{W\text{-}PHY,p}$. It is relevant to the observation-lens analysis shown in Figure 5 and is defined formally in SM12.

This distinction prevents confusion between gene-level aggregation inside one BP observable and the number of BP observables selected for patient-level discriminability scoring.

SM6.12 Output of cohort-level activity–width analysis

The cohort-level activity–width summary step generated the following outputs for each cancer cohort and gene-set database:

$$|\bar{\mu}|_B,$$

the average activity magnitude of each BP observable;

$$\bar{W}_B,$$

the average patient-level observable width of each BP observable;

$$\bar{C}_B,$$

the cohort-level coherence summary of each BP observable;

$$N_B,$$

the mapped gene-set size of each BP observable; and cancer-by-database summary tables containing width and coherence statistics.

These outputs were used to generate Figure 3A–D and served as inputs for downstream uncertainty-aware scoring, random-baseline filtering, and K-scale observation-lens analysis.

SM6.13 Summary

In summary, SM6 defines how patient-level BP activity scores and observable widths were summarized at the cohort level. Each BP observable was represented by average activity magnitude and average observable width:

$$B \Rightarrow (|\bar{\mu}|_B, \bar{W}_B).$$

Coherence was defined as the balance between activity magnitude and observable width:

$$C_{B,p} = \frac{|\mu_{B,p}|}{W_{B,p} + \delta},$$

with cohort-level summaries used for cross-cancer comparison. Gene-set-size analysis was performed using N_B, the number of mapped genes within each BP observable, and was kept distinct from K, the number of selected BP observables used for patient-level score construction. These definitions support the activity–width landscape, cancer-by-database width summary, coherence-based cohort organization, and gene-set scale analysis reported in Figure 3.

SM7 | Uncertainty-aware patient-level score, $D_{W\text{-}PHY,p}(\lambda)$

After patient-level BP activity scores $\mu_{B,p}$ and observable widths $W_{B,p}$ were computed, these quantities were combined to construct an uncertainty-aware patient-level score. The purpose of this score was to evaluate whether process-level transcriptomic activity remains clinically discriminative after accounting for observable-width burden.

This section defines the patient-level activity-magnitude term M_p, the observable-width burden term U_p, and the uncertainty-aware score $D_{W\text{-}PHY,p}(\lambda)$.

SM7.1 Input quantities

For each biological process B and patient p, the following patient-level quantities were available:

$$\mu_{B,p} = \frac{1}{N_B} \sum_{g \in B_{\text{mapped}}} Z_{g,p} \,,$$

and

$$W_{B,p} = \text{FWHM}\big(\{Z_{g,p} : g \in B_{\text{mapped}}\}\big).$$

Here, $\mu_{B,p}$ denotes the signed relative BP activity score, while $W_{B,p}$ denotes the patient-level within-BP observable width.

For patient-level scoring, a selected set of K BP observables was used:

$$\mathcal{B}_K = \{B_1, B_2, \ldots, B_K\}.$$

The definition and selection of K are described in SM12. In the present section, K denotes the number of BP observables included in the patient-level uncertainty-aware score.

SM7.2 Patient-level activity-magnitude term M_p

For each patient p, the activity-magnitude term was defined as the average absolute BP activity score across the selected K BP observables:

$$M_p = \frac{1}{K} \sum_{B \in \mathcal{B}_K} | \mu_{B,p} |.$$

Because $\mu_{B,p}$ is a signed relative activity score, the absolute value was used to quantify deviation magnitude from the cohort-level reference regardless of direction.

Thus, M_p summarizes the average strength of process-level transcriptomic deviation for patient $p$across the selected BP observables.

A larger M_p indicates that the selected BP observables show stronger relative transcriptomic deviation in that patient. A smaller M_p indicates weaker average deviation.

SM7.3 Patient-level observable-width burden U_p

For each patient p, the observable-width burden was defined as the average log-transformed width across the selected KBP observables:

$$U_p = \frac{1}{K} \sum_{B \in \mathcal{B}_K} \log\left(W_{B,p} + \delta\right).$$

Here, $W_{B,p}$is the patient-level FWHM-defined observable width of BP B, and $\delta = 10^{-6}$ is a small numerical stabilization constant.

The logarithmic transformation was used to reduce the influence of extremely broad width values while preserving the ordering of width burden. Thus, U_p summarizes the average observable-width burden of the selected BP observables in patient p.

A larger U_p indicates broader average observable width across the selected BP observables. A smaller U_p indicates more compact average observable width.

SM7.4 Definition of the uncertainty-aware patient-level score

The uncertainty-aware patient-level score was defined as:

$$D_{W\text{-}PHY,p}(\lambda) = M_p - \lambda U_p,$$

where λ controls the strength of observable-width penalization.

Substituting the definitions of M_p and U_p:

$$D_{W\text{-}PHY,p}(\lambda) = \frac{1}{K}\sum_{B\in\mathcal{B}_K} |\mu_{B,p}| - \lambda\left[\frac{1}{K}\sum_{B\in\mathcal{B}_K} \log\,(W_{B,p} + \delta)\right].$$

This formulation integrates process-level activity magnitude and observable-width burden into a single patient-level score.

SM7.5 Activity-only baseline

The special case $\lambda = 0$ corresponds to the activity-only baseline:

$$D_{W\text{-}PHY,p}(0) = M_p.$$

Thus, when $\lambda = 0$, observable width does not contribute to the patient-level score.

This baseline was used to test whether adding observable-width information changed survival-based outcome separation compared with activity magnitude alone. The main manuscript uses this activity-only baseline when interpreting the λ-scan results in Figure 2.

SM7.6 Interpretation of $\boldsymbol{\lambda}$

The parameter λ controls how strongly observable-width burden contributes to the patient-level score.

When:

$$\lambda = 0,$$

the score is based only on activity magnitude.

When:

$$\lambda > 0,$$

patients with broader observable-width burden receive a lower uncertainty-aware score relative to patients with comparable activity magnitude but narrower observable width.

Thus, increasing λ increases the penalty associated with broader process-level observable clouds.

Importantly, λ was treated as an operational weighting parameter, not as a biological constant, physical constant, kinetic parameter, or universal cancer property. Different cancers may show different optimal λ values because observable-width structure and survival association differ across cancer cohorts.

SM7.7 Interpretation of $\boldsymbol{D_{W\text{-}PHY,p}(\lambda)}$

The quantity $D_{W\text{-}PHY,p}(\lambda)$ is a patient-level observable score. It was used to stratify patients for survival analysis.

It should not be interpreted as:

a patient-specific clinical discriminability value,

a direct latent biological-state estimate,

a disease-severity score,

a mechanistic progression score,

or

a physical entropy measure.

Instead, $D_{W\text{-}PHY,p}(\lambda)$ evaluates whether patient-level process activity remains informative after accounting for observable-width burden.

The final survival-based discriminability value is computed later from the log-rank test:

$$D_{\text{clinical}} = -\log_{10}(p_{\text{logrank}}),$$

as described in SM8.

This distinction is important because the main manuscript uses $D_{W\text{-}PHY,p}$ as a patient-level stratification score, whereas D_{clinical} is the outcome-separation metric.

SM7.8 Relationship to coherence

The uncertainty-aware score and coherence are related but not identical.

Patient-level coherence was defined in SM6 as:

$$C_{B,p} = \frac{|\,\mu_{B,p}\,|}{W_{B,p} + \delta}.$$

This is a BP-level compactness measure for one BP observable in one patient.

In contrast, $D_{W\text{-}PHY,p}(\lambda)$ aggregates activity magnitude and width burden across a selected set of K BP observables:

$$D_{W\text{-}PHY,p}(\lambda) = M_p - \lambda U_p.$$

Thus, coherence evaluates the compactness of individual BP observables, whereas $D_{W\text{-}PHY,p}(\lambda)$ constructs a patient-level score from multiple selected BP observables.

Both quantities use activity and width information, but they serve different roles:

$C_{B,p} \rightarrow$ BP-level compactness,

$D_{W\text{-}PHY,p}(\lambda) \rightarrow$ patient-level stratification score.

SM7.9 Construction of patient-level score vectors

For each cancer cohort, selected BP observable set $\mathcal{B}_K$, and λvalue, a patient-level score vector was generated:

$$\mathbf{d}^{(\lambda)} = \big(D_{W\text{-}PHY,1}(\lambda), D_{W\text{-}PHY,2}(\lambda), \dots, D_{W\text{-}PHY,N}(\lambda)\big).$$

Here, Ndenotes the number of matched patients in the cancer cohort.

This score vector was used as the input for survival stratification in SM8. Patients were divided into high-score and low-score groups using the median value of $\mathbf{d}^{(\lambda)}$, and outcome separation was evaluated using Kaplan–Meier and log-rank analysis.

SM7.10 Output of uncertainty-aware patient-level scoring

The uncertainty-aware scoring step generated the following outputs for each cancer cohort, selected BP observable set, and λvalue:

$$M_p,$$

the activity-magnitude term for each patient;

$$U_p,$$

the observable-width burden term for each patient;

$$D_{W\text{-}PHY,p}(\lambda),$$

the uncertainty-aware patient-level score;

and

$$\mathbf{d}^{(\lambda)},$$

the patient-level score vector used for survival stratification.

These outputs were used in SM8 for survival-based clinical discriminability analysis, in SM9 for λ-scan analysis, and in SM12 for K-scale observation-lens analysis.

SM7.11 Summary

In summary, SM7 defines the uncertainty-aware patient-level score:

$$D_{W\text{-}PHY,p}(\lambda) = M_p - \lambda U_p,$$

where:

$$M_p = \frac{1}{K} \sum_{B \in \mathcal{B}_K} | \mu_{B,p} |,$$

and

$$U_p = \frac{1}{K} \sum_{B \in \mathcal{B}_K} \log (W_{B,p} + \delta).$$

The $\lambda = 0$case serves as the activity-only baseline. The resulting $D_{W\text{-}PHY,p}(\lambda)$is a patient-level stratification score, not the final clinical discriminability value. Survival-based clinical discriminability is computed from Kaplan–Meier and log-rank analysis in SM8.

SM8 | Survival-based clinical discriminability

Patient-level uncertainty-aware scores were evaluated using survival-based clinical discriminability analysis. The purpose of this step was to determine whether a given patient-level observable score could separate patients into groups with different clinical outcomes.

This section defines patient stratification, Kaplan–Meier survival estimation, log-rank testing, and the transformation of survival-separation significance into clinical discriminability.

SM8.1 Input patient-level score

For each cancer cohort, selected BP observable set, and uncertainty-weighting value λ, the patient-level uncertainty-aware score was computed as described in SM7:

$$D_{W\text{-}PHY,p}(\lambda) = M_p - \lambda U_p.$$

For a cohort with N matched patients, this generated a score vector:

$$\mathbf{d}^{(\lambda)} = \left(D_{W\text{-}PHY,1}(\lambda), D_{W\text{-}PHY,2}(\lambda), \dots, D_{W\text{-}PHY,N}(\lambda)\right).$$

This score vector was used only for patient stratification. It was not itself interpreted as a survival probability, disease-severity score, latent-state estimate, or final discriminability value.

SM8.2 Median-based patient stratification

Patients were stratified into high-score and low-score groups using the median value of the patient-level score vector:

$$m_\lambda = \mathrm{median}\left(\mathbf{d}^{(\lambda)}\right).$$

The high-score group was defined as:

$$G_{\mathrm{high}}^{(\lambda)} = \left\{p : D_{W\text{-}PHY,p}(\lambda) > m_\lambda\right\},$$

and the low-score group was defined as:

$$G_{\mathrm{low}}^{(\lambda)} = \left\{p : D_{W\text{-}PHY,p}(\lambda) \le m_\lambda\right\}.$$

Median split was used because it provides a consistent and balanced stratification rule across cancer cohorts, pathway collections, λ values, and K-scale analyses. It also avoids arbitrary tuning of cohort-specific thresholds.

SM8.3 Survival data used for evaluation

For each matched patient p, the survival annotation table provided:

$$(t_p, e_p),$$

where t_p denotes survival time and e_p denotes binary event status.

Event status was encoded as:

$$e_p = 0 \text{ for censored observations,}$$

$$e_p = 1 \text{ for observed events.}$$

Overall survival was prioritized as the main clinical endpoint when available. Samples without valid survival time or event-status information were excluded before survival analysis.

SM8.4 Kaplan–Meier survival estimation

For each stratified group, survival probability was estimated using the Kaplan–Meier estimator:

$$\hat{S}(t) = \prod_{t_i \le t} \left(1 - \frac{d_i}{n_i}\right),$$

where d_i denotes the number of observed events at time t_i, and n_i denotes the number of patients at risk immediately before time t_i.

Kaplan–Meier survival curves were estimated separately for:

$$G_{\text{high}}^{(\lambda)}$$

and

$$G_{\text{low}}^{(\lambda)}.$$

These curves were used to visualize whether the patient-level observable score separated patients into groups with different survival trajectories.

SM8.5 Log-rank test

Survival differences between the high-score and low-score groups were evaluated using the log-rank test.

For each λ, the log-rank test compared the survival distributions of:

$$G_{\text{high}}^{(\lambda)} \text{and} G_{\text{low}}^{(\lambda)}.$$

The resulting *p*-value was denoted as:

$$p_{\text{logrank}}(\lambda).$$

A smaller p_{logrank} indicates stronger evidence that the two stratified groups have different survival distributions.

SM8.6 Clinical discriminability score

Clinical discriminability was defined by transforming the log-rank p-value into a positive score:

$$D_{\text{clinical}}(\lambda) = -\log_{10}\left(p_{\text{logrank}}(\lambda)\right).$$

For numerical stability, very small p-values were clipped before transformation:

$$p_{\text{logrank}} \geq 10^{-300}.$$

Thus, the implemented form was:

$$D_{\text{clinical}}(\lambda) = -\log_{10}\left[\max\left(p_{\text{logrank}}(\lambda), 10^{-300}\right)\right].$$

Larger D_{clinical} values indicate stronger survival separation between the high-score and low-score groups.

This definition follows the main manuscript, where uncertainty-aware scores were evaluated by Kaplan–Meier analysis and log-rank testing, and clinical discriminability was defined as $-\log_{10}(p_{\text{logrank}})$.

SM8.7 Distinction between patient-level score and clinical discriminability

The patient-level score and clinical discriminability have different meanings.

The patient-level score:

$$D_{W\text{-}PHY,p}(\lambda)$$

is computed for each patient and is used to stratify patients.

The clinical discriminability value:

$$D_{\text{clinical}}(\lambda)$$

is computed for a stratification result and measures survival separation between the high-score and low-score groups.

Thus:

$$D_{W\text{-}PHY,p}(\lambda) \rightarrow \text{patient-level observable score,}$$

whereas:

$$D_{\text{clinical}}(\lambda) \rightarrow \text{group-level survival discriminability.}$$

This distinction is essential. A patient does not have an individual clinical discriminability value. Clinical discriminability is obtained only after patient scores are used to form survival groups and those groups are evaluated by log-rank testing.

SM8.8 Activity-only baseline discriminability

For the activity-only baseline, $\lambda = 0$, the patient-level score reduces to:

$$D_{W\text{-}PHY,p}(0) = M_p.$$

The corresponding baseline clinical discriminability was:

$$D_{\text{clinical}}(0) = -\log_{10}\left(p_{\text{logrank}}[D_{W\text{-}PHY,p}(0)]\right).$$

This baseline was used to evaluate whether incorporating observable width altered survival-based discriminability relative to activity magnitude alone.

In the main manuscript, this baseline is represented by the $\lambda = 0$ condition in the λ-scan analysis shown in Figure 2.

SM8.9 Quality-control criteria for survival analysis

To reduce unstable survival estimates, survival analysis was skipped when the available data did not satisfy minimum quality-control criteria.

A survival comparison was excluded if:

$$N < 30,$$

where N is the number of matched patients;

$$E < 5,$$

where E is the total number of observed events; or if either median-split group contained too few patients:

$$n_{\text{group}} < 10.$$

These criteria were used to prevent unstable Kaplan–Meier curves and unreliable log-rank p-values caused by sparse event counts or very small stratified groups.

SM8.10 Handling ties at the median

When patient-level scores contained ties at the median, the stratification rule was applied consistently:

$$D_{W\text{-}PHY,p}(\lambda) > m_\lambda \rightarrow G_{\text{high}}^{(\lambda)},$$

$$D_{W\text{-}PHY,p}(\lambda) \leq m_\lambda \rightarrow G_{\text{low}}^{(\lambda)}.$$

This rule ensured deterministic group assignment across all cancer cohorts, λ values, and K-scale analyses.

SM8.11 Interpretation of $\boldsymbol{D}_{\textbf{clinical}}$

D_{clinical} was interpreted as an operational measure of outcome separation associated with a patient-level observable score.

Larger values indicate stronger survival separation, but they do not imply:

causal biological mechanism,

direct latent-state reconstruction,

therapeutic response prediction,

or

universal clinical validity.

Instead, D_{clinical} quantifies how well the tested observable separates patients with different survival outcomes under the current cohort, endpoint, stratification rule, and analysis pipeline.

This interpretation is important because the main manuscript argues that high discriminability alone is insufficient for biological interpretation. Discriminability must be evaluated together with observable width, coherence, random-background behavior, biological overlap, and reproducibility.

SM8.12 Output of survival-based discriminability analysis

For each cancer cohort, selected BP observable set, and λvalue, the survival-based discriminability analysis generated:

$$G_{\text{high}}^{(\lambda)},$$

the high-score patient group;

$$G_{\text{low}}^{(\lambda)},$$

the low-score patient group;

$$p_{\text{logrank}}(\lambda),$$

the log-rank p-value;

$$D_{\text{clinical}}(\lambda),$$

the survival-based clinical discriminability score; and Kaplan–Meier survival-curve outputs for visualization when required.

These outputs were used in SM9 for λ-scan analysis, in SM10–SM11 for random-baseline and *D* + coherence filtering, and in SM12 for K-scale observation-lens analysis.

SM8.13 Summary

In summary, SM8 defines how patient-level uncertainty-aware scores were converted into survival-based clinical discriminability. Patients were stratified by median split using $D_{W\text{-}PHY,p}(\lambda)$, Kaplan–Meier curves were estimated for high-score and low-score groups, and survival separation was evaluated using the log-rank test. Clinical discriminability was defined as:

$$D_{\text{clinical}}(\lambda) = -\log_{10}\big(p_{\text{logrank}}(\lambda)\big).$$

The patient-level score $D_{W\text{-}PHY,p}(\lambda)$ is therefore a stratification score, whereas $D_{\text{clinical}}(\lambda)$ is the outcome-separation metric used to evaluate discriminability.

SM9 | λ-scan analysis and uncertainty-sensitivity metrics

After defining the uncertainty-aware patient-level score and survival-based clinical discriminability, a λ-scan analysis was performed to evaluate whether observable-width weighting altered clinical discriminability relative to activity-only scoring. The purpose of this analysis was to test whether patient-level observable width $W_{B,p}$ contributed outcome-relevant information beyond BP activity magnitude alone.

This section defines the λ-grid, activity-only baseline, optimal λ, discriminability shift, and uncertainty-sensitivity amplitude used to summarize cancer-specific responses to observable-width weighting.

SM9.1 Input from uncertainty-aware scoring and survival analysis

For each cancer cohort, a patient-level uncertainty-aware score was computed for each value of λ:

$$D_{W\text{-}PHY,p}(\lambda) = M_p - \lambda U_p,$$

where:

$$M_p = \frac{1}{K} \sum_{B \in \mathcal{B}_K} \mid \mu_{B,p} \mid,$$

and:

$$U_p = \frac{1}{K} \sum_{B \in \mathcal{B}_K} \log\left(W_{B,p} + \delta\right).$$

For each λ, patients were stratified by median split, and survival-based clinical discriminability was computed as:

$$D_{\text{clinical}}(\lambda) = -\log_{10}\left(p_{\text{logrank}}(\lambda)\right).$$

Thus, the λ-scan generated a cancer-specific discriminability response curve:

$$D_{\text{clinical}}(\lambda).$$

SM9.2 λ-grid

The uncertainty-weighting parameter $\lambda$was scanned over a fixed grid:

$$\lambda \in \{0,\ 0.25,\ 0.50,\ 0.75,\ 1.00,\ 1.50,\ 2.00\}.$$

The same λ-grid was used across all cancer cohorts to enable consistent cross-cancer comparison.

The value:

$$\lambda = 0$$

corresponds to the activity-only baseline, whereas:

$$\lambda > 0$$

introduces increasing penalization of observable-width burden.

SM9.3 Activity-only baseline

At $\lambda = 0$, the uncertainty-aware patient-level score reduces to:

$$D_{W\text{-}PHY,p}(0) = M_p.$$

The corresponding clinical discriminability was defined as:

$$D_{\text{baseline}} = D_{\text{clinical}}(0).$$

This baseline measures the survival-separation ability of activity magnitude alone, without incorporating observable-width burden.

The λ-scan was therefore interpreted relative to this baseline. If observable width did not contribute outcome-relevant information, then $D_{\text{clinical}}(\lambda)$ would be expected to remain close to $D_{\text{clinical}}(0)$ across the λ-grid. In contrast, deviations from the baseline indicate that observable-width weighting changes survival-based discriminability.

This is the logic used in the main manuscript to interpret Figure 2, where dashed horizontal lines indicate the $\lambda = 0$ activity-only baseline.

SM9.4 Optimal uncertainty weighting

For each cancer cohort, the optimal uncertainty-weighting value was defined as the λvalue that produced the maximum clinical discriminability:

$$\lambda^* = \arg\max_{\lambda} D_{\text{clinical}}(\lambda).$$

The corresponding maximum discriminability was:

$$D_{\max} = \max_{\lambda} D_{\text{clinical}}(\lambda).$$

The pair:

$$(\lambda^{*}, D_{\max})$$

summarizes the peak uncertainty-weighted discriminability response for each cancer cohort.

Because λ^{*} was selected from the fixed grid, it should be interpreted as the best-performing tested uncertainty-weighting value rather than as a continuous optimized physical or biological parameter.

SM9.5 Discriminability shift relative to activity-only baseline

To quantify how much observable-width weighting changed discriminability relative to activity-only scoring, the discriminability shift was defined as:

$$\Delta D_{\text{baseline}} = D_{\max} - D_{\text{baseline}}.$$

Equivalently:

$$\Delta D_{\text{baseline}} = \max_{\lambda} D_{\text{clinical}}(\lambda) - D_{\text{clinical}}(0).$$

A positive value indicates that at least one $\lambda > 0$ or tested λ condition improved survival-based discriminability relative to the activity-only baseline. A value close to zero indicates little improvement relative to activity-only scoring. A negative value is generally not expected when $D_{\max}$ is defined over a grid that includes $\lambda = 0$, because the maximum includes the baseline condition. However, individual nonzero λ values may show lower discriminability than the activity-only baseline.

This metric supports Figure 2C, where discriminability shifts relative to the activity-only baseline were summarized across cancer cohorts.

SM9.6 Uncertainty-sensitivity amplitude

To quantify how strongly discriminability varied across uncertainty-weighting values, the uncertainty-sensitivity amplitude was defined as the range of discriminability values across the λ-grid:

$$A_\lambda = \max_\lambda D_{\text{clinical}}(\lambda) - \min_\lambda D_{\text{clinical}}(\lambda).$$

A larger A_λ indicates that survival-based discriminability is more sensitive to observable-width weighting. A smaller A_λ indicates that discriminability is relatively stable across λ, suggesting weaker dependence on the width-penalty term.

This metric supports Figure 2D, where uncertainty-sensitivity amplitude was compared across cancer cohorts.

SM9.7 Peak-response region

For visualization, the peak-response region was identified around the λ value producing maximal clinical discriminability:

$$\lambda^* = \arg\max_\lambda D_{\text{clinical}}(\lambda).$$

In Figure 2A, vertical dashed markers indicate peak-response weighting regions for representative cancer-specific λ-response curves. These markers were used for visual interpretation only and did not imply that λ^* is a mechanistic cancer parameter.

SM9.8 Cancer-specific λ-response profiles

For each cancer cohort, the λ-scan generated a response profile:

$$\mathcal{R}_c = \left\{ D_{\mathrm{clinical}}^{(c)}(\lambda) : \lambda \in \Lambda \right\},$$

where c denotes the cancer cohort and Λ denotes the fixed λ-grid.

Cancer-specific response profiles were compared according to:

$$D_{\mathrm{baseline}},$$
$$D_{\max},$$
$$\lambda^*,$$
$$\Delta D_{\mathrm{baseline}},$$

and:

$$A_\lambda.$$

Different cancer cohorts may show different response profiles because observable-width burden, activity structure, survival endpoint informativeness, and cohort-specific transcriptomic organization vary across cancers.

This interpretation is consistent with the main manuscript, which states that distinct cancer types exhibited different response profiles as λ varied, indicating cancer-dependent contributions of observable width to outcome separation.

SM9.9 Interpretation of λ-scan results

The λ-scan was designed to test whether incorporating patient-level observable width changes survival-based discriminability.

The interpretation was as follows:

If:

$$D_{\mathrm{clinical}}(\lambda) \approx D_{\mathrm{clinical}}(0) \text{ for all } \lambda,$$

then activity magnitude alone provides most of the discriminative information under the tested score construction.

If:

$$D_{\text{clinical}}(\lambda) > D_{\text{clinical}}(0) \text{for some } \lambda,$$

then observable-width weighting improves clinical discriminability relative to activity-only scoring.

If:

$$D_{\text{clinical}}(\lambda) < D_{\text{clinical}}(0) \text{for some } \lambda,$$

then excessive or inappropriate width penalization may suppress outcome-relevant activity information.

Therefore, λ-scan analysis was not interpreted as proof that width universally improves discriminability. Instead, it was interpreted as a diagnostic test of whether and how observable-width structure modifies outcome separation in a cancer-dependent manner.

This interpretation follows the main Discussion, which emphasizes that the central contribution is not that W universally improves discriminability, but that W changes how discriminability should be interpreted.

SM9.10 Relationship to Figure 2

The λ-scan analysis directly supports Figure 2 of the main manuscript.

Figure 2A used cancer-specific curves:

$$D_{\text{clinical}}(\lambda)$$

to show how discriminability changed across uncertainty-weighting values.

Figure 2B summarized cancer-specific peak-response weighting values:

$$\lambda^*.$$

Figure 2C summarized discriminability shifts relative to the activity-only baseline:

$$\Delta D_{\text{baseline}} = D_{\max} - D_{\text{clinical}}(0).$$

Figure 2D summarized uncertainty-sensitivity amplitude:

$$A_\lambda = \max_\lambda D_{\text{clinical}}(\lambda) - \min_\lambda D_{\text{clinical}}(\lambda).$$

Together, these panels evaluate whether observable-width weighting affects survival-based discriminability and whether this effect differs across cancer cohorts.

SM9.11 Output of λ-scan analysis

For each cancer cohort, the λ-scan generated the following outputs:

$$D_{\text{clinical}}(\lambda) \text{for each tested } \lambda,$$
$$D_{\text{baseline}} = D_{\text{clinical}}(0),$$
$$\lambda^*,$$
$$D_{\max},$$
$$\Delta D_{\text{baseline}},$$

and:

$$A_\lambda.$$

These outputs were exported as summary tables and used for Figure 2 construction, cross-cancer comparison, and downstream observation-scale analysis.

SM9.12 Summary

In summary, SM9 defines the λ-scan analysis used to evaluate cancer-specific sensitivity to observable-width weighting. For each λ, patients were scored using:

$$D_{W\text{-}PHY,p}(\lambda) = M_p - \lambda U_p,$$

stratified by median split, and evaluated using:

$$D_{\text{clinical}}(\lambda) = -\log_{10}\big(p_{\text{logrank}}(\lambda)\big).$$

The activity-only baseline was defined at $\lambda = 0$. The optimal tested uncertainty weighting λ^*, maximum discriminability $D_{\max}$, baseline shift $\Delta D_{\text{baseline}}$, and uncertainty-sensitivity amplitude A_λ were used to summarize cancer-specific λ-response behavior. This analysis supports Figure 2 and provides the methodological basis for interpreting how observable width modifies survival-based discriminability across cancer cohorts.

SM10 | Size-matched random baseline generation

Size-matched random gene-set baselines were generated to evaluate whether the discriminability of structured biological-process observables exceeded the level expected from random aggregation of transcriptomic genes. The purpose of this analysis was not to assume that random gene sets are biologically meaningless, but to provide an empirical background for interpreting high-discriminability observables.

This section defines random gene-set generation, size matching, random discriminability distributions, random 95th percentile thresholds, and structured BP fractions exceeding random-background expectation.

SM10.1 Rationale for random-baseline analysis

High-dimensional transcriptomic data can produce outcome-associated signals even from randomly selected gene sets. Therefore, a high survival-based discriminability value does not automatically imply that a biological-process observable is mechanistically meaningful, biologically interpretable, or structurally reliable.

To address this issue, structured BP observables were compared against size-matched random gene sets generated from the same cohort-specific transcriptomic gene universe. This random-baseline analysis was used to evaluate whether structured biological-process observables exceeded empirical random-background expectation under the same preprocessing, scoring, and survival-analysis pipeline.

This rationale directly supports the main manuscript's Figure 4 logic, where random baselines were used to show that discriminability alone is insufficient and should be interpreted together with width-aware coherence and biological structure.

SM10.2 Input gene universe

For each cancer cohort, the available transcriptomic gene universe was defined from the processed and z-score-normalized expression matrix:

$$G_{\mathrm{expr}} = \{g_1, g_2, \dots, g_G\}.$$

Only genes present in the processed expression matrix were eligible for random gene-set construction.

This ensured that random gene sets and structured BP gene sets were sampled from the same measurable gene universe.

SM10.3 Structured BP observables used for random comparison

For each structured BP observable B, the mapped gene set was defined as:

$$B_{\mathrm{mapped}} = B \cap G_{\mathrm{expr}},$$

with mapped gene-set size:

$$N_B = |B_{\mathrm{mapped}}|.$$

Random-baseline comparison was performed on structured BP observables satisfying the controlled mapped-gene-size range:

$$10 \le N_B \le 300.$$

This controlled range was used to reduce instability from very small gene sets and excessive signal dilution from overly broad gene sets. It also ensured that structured and random observables were compared within a reasonable gene-set aggregation scope.

SM10.4 Size-matched random gene-set generation

For each structured BP observable B, random gene sets were generated by sampling N_B genes from the cohort-specific expression gene universe:

$$R_{B,j} = \{r_1, r_2, \dots, r_{N_B}\},$$

where:

$$R_{B,j} \subseteq G_{\text{expr}},$$

and:

$$| R_{B,j} |= N_B.$$

Here, j indexes the random repetition for structured observable B.

Thus, every random gene set was size-matched to its corresponding structured BP observable. This size matching was necessary because gene-set size affects activity averaging, observable width estimation, and survival-based discriminability. The mapped gene-set size N_B was therefore preserved when constructing the empirical random baseline.

SM10.5 Number of random repetitions

For each structured BP observable, multiple independent size-matched random gene sets were generated:

$$j = 1,2, \dots, N_{\text{random}}.$$

In the implemented workflow:

$$N_{\text{random}} = 20.$$

Thus, each retained structured BP observable was associated with 20 size-matched random controls.

The resulting collection of random gene sets was used to estimate the empirical random-discriminability distribution for each cancer cohort and pathway database.

SM10.6 Processing of random gene sets

Each random gene set $R_{B,j}$ was processed using the same workflow applied to structured BP observables.

For each patient p, the random observable cloud was defined as:

$$O_{R_{B,j},p} = \{Z_{g,p}: g \in R_{B,j}\}.$$

The random activity score was computed as:

$$\mu_{R_{B,j},p} = \frac{1}{N_B} \sum_{g \in R_{B,j}} Z_{g,p} .$$

The random observable width was estimated as:

$$W_{R_{B,j},p} = \mathrm{FWHM}\left(O_{R_{B,j},p}\right).$$

Thus, random gene sets were evaluated using the same patient-level activity and patient-level width definitions as structured BP observables.

SM10.7 Random observable discriminability

For each random gene set $R_{B,j}$, patient-level activity and width quantities were used to construct patient-level scores and evaluate survival-based clinical discriminability using the same procedure described in SM7–SM9.

The clinical discriminability of each random gene set was defined as:

$$D_{\mathrm{random}}(R_{B,j}) = -\log_{10}\left(p_{\mathrm{logrank}}(R_{B,j})\right).$$

Here, $p_{\text{logrank}}(R_{B,j})$ denotes the log-rank p-value obtained after stratifying patients using the random-set-derived score.

The same median-split rule, survival endpoint, log-rank test, p-value clipping, and group-size quality-control thresholds were used for structured and random observables.

SM10.8 Random discriminability distribution

For each cancer cohort and gene-set database, random discriminability values were pooled to form an empirical random-background distribution:

$$\mathcal{D}_{\text{random}} = \{D_{\text{random}}(R_{B,j}) : B \in \mathcal{B}_{\text{structured}}, j = 1, \dots, N_{\text{random}}\}.$$

This empirical distribution represents the discriminability expected from size-matched random gene aggregation under the same transcriptomic cohort, endpoint, preprocessing, scoring, and survival-analysis workflow.

From this distribution, the following summary statistics were computed:

$$\bar{D}_{\text{random}},$$

the mean random discriminability;

$$D_{\text{random}}^{\max},$$

the maximum random discriminability; and:

$$Q_{95}^{\text{random}},$$

the 95th percentile of the random discriminability distribution.

SM10.9 Random 95th percentile threshold

The random 95th percentile threshold was defined as:

$$Q_{95}^{\text{random}} = \text{Percentile}_{95}(\mathcal{D}_{\text{random}}).$$

This threshold was used as an empirical random-background reference. A structured BP observable was considered to exceed random-background expectation when:

$$D_{\text{structured}}(B) > Q_{95}^{\text{random}}.$$

This threshold does not imply definitive biological significance. Instead, it indicates that a structured BP observable shows higher discriminability than most size-matched random observables under the same analysis conditions.

SM10.10 Structured BP discriminability distribution

For each structured BP observable B, clinical discriminability was computed using the same survival-based procedure:

$$D_{\text{structured}}(B) = -\log_{10}\big(p_{\text{logrank}}(B)\big).$$

The structured discriminability distribution for a cancer cohort and database was defined as:

$$\mathcal{D}_{\text{structured}} = \{D_{\text{structured}}(B) : B \in \mathcal{B}_{\text{structured}}\}.$$

From this distribution, the following summary statistics were computed:

$$\bar{D}_{\text{structured}},$$
$$D_{\text{structured}}^{\max},$$

and the number or fraction of structured BP observables exceeding random-background expectation.

SM10.11 Fraction of structured BP observables exceeding random background

For each cancer cohort and gene-set database, the fraction of structured BP observables exceeding the random 95th percentile was defined as:

$$F_{\text{above}} = \frac{\#\{B : D_{\text{structured}}(B) > Q_{95}^{\text{random}}\}}{N_{\text{structured}}},$$

where $N_{\text{structured}}$ denotes the total number of structured BP observables evaluated in that cancer–database combination.

This fraction was used to quantify how often structured BP observables exceeded the empirical random-background threshold.

This metric supports Figure 4A of the main manuscript, where the fraction of structured BP observables above the random 95th percentile was summarized across cancer cohorts and pathway databases.

SM10.12 Structured maximum D versus random maximum D

To evaluate whether the strongest structured BP observables clearly exceeded the strongest random projections, the maximum structured and random discriminability values were compared:

$$D_{\text{structured}}^{\max} = \max_{B} \ D_{\text{structured}}(B),$$

and:

$$D_{\text{random}}^{\max} = \max_{R} \ D_{\text{random}}(R).$$

This comparison was used to detect cases where random gene sets achieved discriminability comparable to or greater than structured BP observables.

This metric supports Figure 4B of the main manuscript. The main text uses this result to emphasize that maximum discriminability alone is insufficient for biological interpretation, because random gene sets can occasionally achieve high D.

SM10.13 Interpretation of random-baseline results

The random-baseline analysis was interpreted as a diagnostic framework rather than a simple pass/fail statistical test.

If many structured BP observables exceeded:

$$Q_{95}^{\text{random}},$$

then structured biological-process organization was considered enriched for discriminability above empirical random-background expectation.

If random observables achieved high:

$$D_{\text{random}}^{\max},$$

then high discriminability alone was considered insufficient for biological interpretation.

Thus, random-baseline results were used to separate:

outcome separation alone

from:

outcome separation supported by biological structure and coherence.

This distinction is central to the main manuscript's Figure 4 interpretation: random baselines reveal why Dshould be used as a screening criterion rather than as a final interpretation.

SM10.14 Relationship to downstream D + coherence filtering

The random 95th percentile threshold:

$$Q_{95}^{\text{random}}$$

was used as the first filter in the downstream D + coherence analysis described in SM11.

A structured BP observable was first required to exceed the random-background threshold:

$$D_{\text{structured}}(B) > Q_{95}^{\text{random}}.$$

It was then evaluated for width-aware coherence. This two-stage filtering approach was used to identify structured BP observables that showed both outcome separation and compact activity–width organization.

SM10.15 Output of random-baseline generation

The random-baseline workflow generated the following outputs for each cancer cohort and gene-set database:

$$\mathcal{D}_{\mathrm{random}},$$

the empirical random discriminability distribution;

$$Q_{95}^{\mathrm{random}},$$

the random 95th percentile threshold;

$$D_{\mathrm{random}}^{\max},$$

the maximum random discriminability;

$$\mathcal{D}_{\mathrm{structured}},$$

the structured BP discriminability distribution;

$$D_{\mathrm{structured}}^{\max},$$

the maximum structured BP discriminability; and:

$$F_{\mathrm{above}},$$

the fraction of structured BP observables exceeding random-background expectation.

These outputs were used to generate Figure 4A and Figure 4B and served as inputs for the D + coherence filter and biological-overlap analysis described in SM11.

SM10.16 Summary

In summary, SM10 defines the size-matched random-baseline workflow used to evaluate structured BP observables against empirical random-background expectation. For each structured BP observable B, random gene sets of identical mapped size $N_B$were sampled from the cohort-specific gene universe. Random and structured observables were processed using identical activity, width, scoring, and survival-discriminability procedures.

The random 95th percentile threshold:

$$Q_{95}^{\mathrm{random}}$$

was used to identify structured BP observables exceeding random-background expectation, and the fraction:

$$F_{\text{above}} = \frac{\#(D_{\text{structured}} > Q_{95}^{\text{random}})}{N_{\text{structured}}}$$

was used to summarize enrichment above random background. Comparison of maximum structured and random discriminability values further tested whether high D alone was sufficient for interpretation. These analyses support the first two panels of Figure 4 and provide the random-background foundation for the D + coherence and biological-overlap analyses in SM11.

SM11 | *D* + coherence filtering and biological-overlap analysis

After size-matched random-baseline analysis, structured BP observables were further evaluated using width-aware coherence and biological-overlap diagnostics. The purpose of this analysis was to distinguish high-discriminability observables that also exhibited compact activity–width organization from high-discriminability signals that may arise from less interpretable random projections.

This section defines the *D* + coherence filter used for structured BP observables and the biological-overlap analysis used to evaluate high-discriminability random gene sets.

SM11.1 Rationale for D + coherence filtering

Clinical discriminability alone is insufficient for biological interpretation because random gene sets can occasionally achieve high survival separation. Therefore, a structured BP observable was not interpreted as informative solely because it had a high D_{clinical} value.

Instead, the present analysis evaluated whether a structured BP observable satisfied two conditions:

1. It exceeded random-background discriminability expectation.
2. It exhibited sufficient width-aware coherence.

This two-condition framework was designed to identify BP observables that showed both outcome separation and compact activity–width organization. The main manuscript uses this logic in Figure 4C, where structured BP observables passing both a random-background D threshold and a coherence filter were summarized.

SM11.2 Input quantities

For each structured BP observable B, the following quantities were available:

$$D_{\text{structured}}(B),$$

the survival-based clinical discriminability of B;

$$\bar{C}_B,$$

the cohort-level coherence summary of B; and:

$$Q_{95}^{\text{random}},$$

the 95th percentile of the size-matched random discriminability distribution.

The discriminability value $D_{\text{structured}}(B)$ was computed using the survival-based procedure described in SM8, while $\bar{C}_B$ was computed from patient-level activity and width as described in SM6.

SM11.3 Random-background D threshold

The first filtering criterion required a structured BP observable to exceed the random-background threshold:

$$D_{\text{structured}}(B) > Q_{95}^{\text{random}}.$$

Here,

$$Q_{95}^{\text{random}} = \text{Percentile}_{95}(\mathcal{D}_{\text{random}}),$$

where $\mathcal{D}_{\text{random}}$ is the empirical distribution of size-matched random discriminability values for the corresponding cancer cohort and pathway database.

This criterion identifies structured BP observables whose survival-based discriminability exceeds most size-matched random projections.

SM11.4 Coherence threshold

The second filtering criterion required the structured BP observable to exhibit sufficient width-aware coherence.

For each BP observable B, cohort-level coherence was defined as:

$$\bar{C}_B = \frac{1}{N}\sum_{p=1}^{N}\frac{\mid \mu_{B,p} \mid}{W_{B,p}+\delta}.$$

A coherence threshold was defined within each cancer–database combination. In the implemented analysis, a BP observable was considered coherence-supported if its coherence value exceeded the upper quartile of coherence values among structured BP observables:

$$\bar{C}_B > Q_{75}^{\text{coherence}},$$

where:

$$Q_{75}^{\text{coherence}} = \text{Percentile}_{75}(\{\bar{C}_B : B \in \mathcal{B}_{\text{structured}}\}).$$

This threshold was used as an operational filter for compact activity–width organization. It was not interpreted as a universal biological threshold.

SM11.5 *D* + coherence filter

A structured BP observable was defined as passing the *D* + coherence filter if it satisfied both:

$$D_{\text{structured}}(B) > Q_{95}^{\text{random}},$$

and:

$$\bar{C}_B > Q_{75}^{\text{coherence}}.$$

The set of structured BP observables passing both filters was defined as:

$$\mathcal{B}_{D+C} = \{B : D_{\text{structured}}(B) > Q_{95}^{\text{random}} \ \wedge \ \bar{C}_B > Q_{75}^{\text{coherence}}\}.$$

This filter identifies structured BP observables that show both survival-based outcome separation and compact activity–width structure.

SM11.6 Fraction passing *D* + coherence filter

For each cancer cohort and pathway database, the fraction of structured BP observables passing both filters was defined as:

$$F_{D+C} = \frac{|\ \mathcal{B}_{D+C}\ |}{N_{\text{structured}}},$$

where $N_{\text{structured}}$ denotes the total number of structured BP observables evaluated in the corresponding cancer–database combination.

This fraction was used to generate Figure 4C of the main manuscript. The purpose of Figure 4C is to show that requiring both random-background discriminability and coherence narrows the candidate set relative to discriminability alone.

SM11.7 Interpretation of *D* + coherence filtering

The *D* + coherence filter was interpreted as a diagnostic selection step rather than a definitive biological validation.

A BP observable passing the *D* + coherence filter was considered more interpretable because it satisfied both:

outcome separation above random background,

and:

compact activity–width organization.

However, passing this filter does not prove causality, mechanism, or clinical validity. It only indicates that the observable is less likely to be a purely high-D but diffuse or poorly organized projection.

Thus, *D* + coherence filtering supports the main manuscript's interpretation that informative observables should be evaluated using discriminability, width-aware coherence, random-background expectation, and biological structure rather than D alone.

SM11.8 High-*D* random gene-set extraction

To evaluate whether high-discriminability random gene sets showed overlap with known biological structure, high-*D* random sets were extracted from the random-baseline analysis.

A random gene set R was classified as high-D if:

$$D_{\text{random}}(R) \geq Q_{95}^{\text{random}}.$$

The set of high-*D* random gene sets was defined as:

$$\mathcal{R}_{\text{highD}} = \{R: D_{\text{random}}(R) \geq Q_{95}^{\text{random}}\}.$$

This extraction was used for biological-overlap analysis.

SM11.9 Biological-overlap analysis using Jaccard index

For each high-*D* random gene set R, overlap with structured biological annotations was evaluated using the Jaccard index.

For a random gene set R and a structured BP gene set B, the Jaccard overlap was defined as:

$$J(R, B) = \frac{| R \cap B_{\text{mapped}} |}{| R \cup B_{\text{mapped}} |}.$$

For each high-*D* random gene set R, the maximum Jaccard overlap with any structured BP annotation was computed as:

$$J_{\max}(R) = \max_{B \in \mathcal{B}_{\text{structured}}} J(R, B).$$

This maximum-overlap statistic evaluates whether a high-*D* random gene set shares measurable gene-membership structure with known biological-process annotations.

SM11.10 Summary of high-*D* random biological overlap

For each cancer cohort and pathway database, the distribution of maximum Jaccard overlaps was summarized across high-*D* random gene sets:

$$\mathcal{J}_{\text{highD}} = \{J_{\max}(R) : R \in \mathcal{R}_{\text{highD}}\}.$$

Summary statistics included:

$$\bar{J}_{\max},$$

the mean maximum Jaccard overlap;

$$\text{median}(J_{\max}),$$

the median maximum Jaccard overlap; and the distribution of $J_{\max}$ values across high-*D* random sets.

These outputs were used to generate Figure 4D of the main manuscript. Figure 4D evaluates whether high-*D* random gene sets show overlap with structured biological annotations or remain less interpretable random projections.

SM11.11 Interpretation of biological-overlap results

The biological-overlap analysis was interpreted cautiously.

A high-*D* random gene set with measurable overlap to structured BP annotations may capture fragments of organized transcriptomic variation. However, overlap alone does not prove that the random gene set represents a coherent biological mechanism.

Conversely, a high-*D* random gene set with low overlap may still reflect distributed transcriptomic structure not captured by existing annotations, but such a signal is less directly interpretable.

Therefore, high-*D* random gene sets were treated as diagnostic signals rather than mechanistic findings. This interpretation is consistent with the main manuscript, where random high-*D* signals are described as requiring additional biological-structure checks before interpretation.

SM11.12 Relationship between D, coherence, and biological overlap

The three diagnostic layers used in Figure 4 have different roles:

$$D_{\mathrm{clinical}}$$

evaluates outcome separation.

$$\bar{C}_B$$

evaluates compact activity–width organization.

$$J_{\max}$$

evaluates overlap between high-*D* random sets and structured biological annotations.

Together, these layers distinguish:

high-*D* but weakly interpretable projections

from:

high-*D*, coherent, biologically structured observables.

Thus, the combined diagnostic framework supports the central interpretation that D is a screening metric, not a complete interpretation criterion.

SM11.13 Relationship to Figure 4

SM11 directly supports Figure 4C and Figure 4D.

Figure 4C used:

$$F_{D+C} = \frac{|\,\mathcal{B}_{D+C}\,|}{N_{\mathrm{structured}}}$$

to summarize the fraction of structured BP observables passing both the random-background *D* threshold and coherence filter.

Figure 4D used:

$$J_{\max}(R) = \max_{B} \frac{|\,R \cap B_{\mathrm{mapped}}\,|}{|\,R \cup B_{\mathrm{mapped}}\,|}$$

to summarize the biological overlap of high-*D* random gene sets with structured annotations.

Together with Figure 4A–B, these analyses show why high discriminability must be interpreted with random-background expectation, width-aware coherence, and biological structure.

SM11.14 Output of *D* + coherence and biological-overlap analysis

The *D* + coherence filtering workflow generated the following outputs:

$$Q_{95}^{\mathrm{random}},$$

the random-background discriminability threshold;

$$Q_{75}^{\mathrm{coherence}},$$

the coherence threshold;

$$\mathcal{B}_{D+C},$$

the set of structured BP observables passing both filters;

$$F_{D+C},$$

the fraction of structured BP observables passing both filters.

The biological-overlap workflow generated:

$$\mathcal{R}_{\mathrm{highD}},$$

the set of high-*D* random gene sets;

$$J(R, B),$$

the Jaccard overlap between each high-*D* random set and structured BP annotation;

$$J_{\max}(R),$$

the maximum biological overlap of each high-*D* random set; and:

$$\mathcal{J}_{\mathrm{highD}},$$

the distribution of maximum overlaps across high-*D* random sets.

These outputs were used for Figure 4C–D and for supplementary random-baseline interpretation.

SM11.15 Summary

In summary, SM11 defines the *D* + coherence and biological-overlap analyses used to extend the random-baseline workflow. A structured BP observable was considered more interpretable when it exceeded the random 95th percentile threshold and passed a coherence filter:

$$D_{\mathrm{structured}}(B) > Q_{95}^{\mathrm{random}},$$
$$\bar{C}_B > Q_{75}^{\mathrm{coherence}}.$$

The fraction of structured BP observables passing both conditions was used to support Figure 4C.

High-*D* random gene sets were further evaluated using maximum Jaccard overlap with structured BP annotations:

$$J_{\max}(R) = \max_{B \in \mathcal{B}_{\mathrm{structured}}} \frac{| R \cap B_{\mathrm{mapped}} |}{| R \cup B_{\mathrm{mapped}} |}.$$

This analysis supported Figure 4D and provided a biological-overlap diagnostic for random high-D signals. Together, these analyses support the main manuscript's conclusion that high D alone is insufficient and should be interpreted jointly with random background, width-aware coherence, and biological structure.

SM12 | K-scale observation-lens analysis

K-scale observation-lens analysis was performed to evaluate whether uncertainty-aware clinical discriminability depends on the number of selected biological-process observables used to construct the patient-level score. The purpose of this analysis was to test whether increasing the number of BP observables improves outcome separation monotonically or whether discriminability exhibits scale-dependent behavior.

This analysis supports the scale-dependent results reported in Figure 5 of the main manuscript.

SM12.1 Rationale for K-scale analysis

Biological-process observables are collective transcriptomic measurements. Their usefulness may depend not only on the activity and width of individual BP gene sets, but also on how many BP observables are combined to construct a patient-level score.

A small number of BP observables may capture highly coherent process-level signals but may fail to cover broader outcome-relevant transcriptomic organization. Conversely, using too many BP observables may introduce weak, redundant, broad, or less outcome-relevant signals that dilute discriminability.

Therefore, the analysis tested whether uncertainty-aware discriminability depends on the observable-set scale:

$$K.$$

In this context, K represents the number of selected BP observables used to construct the patient-level uncertainty-aware score. It does not represent the number of genes inside a BP gene set.

SM12.2 Distinction between N_B and K

Two scale quantities were used in this study:

$$N_B = | B_{\text{mapped}} |,$$

where N_B denotes the number of mapped genes within biological process B, and:

$$K = | \mathcal{B}_K |,$$

where K denotes the number of selected BP observables included in the patient-level score.

These two quantities describe different biological and computational levels.

$$N_B$$

describes the gene-set aggregation scope of one BP observable. It determines the size of:

$$O_{B,p} = \{Z_{g,p} : g \in B_{\text{mapped}}\},$$

and is relevant to the gene-set-size analysis in Figure 3D.

$$K$$

describes the number of BP observables combined into the patient-level uncertainty-aware score:

$$D_{W\text{-}PHY,p}(\lambda).$$

It is relevant to the observation-lens analysis in Figure 5.

Thus, Figure 3D evaluates how observable width depends on the number of genes within each BP observable, whereas Figure 5 evaluates how discriminability depends on the number of BP observables selected for patient-level scoring. This distinction is explicitly reflected in the main manuscript's interpretation of gene-set aggregation scope and observable-set scale.

SM12.3 BP observable ranking for *K*-scale analysis

Before constructing K-dependent patient-level scores, BP observables were ranked according to coherence.

For each BP observable B, cohort-level coherence was defined as:

$$\bar{C}_B = \frac{1}{N} \sum_{p=1}^{N} \frac{|\mu_{B,p}|}{W_{B,p} + \delta}.$$

BP observables were sorted in descending order of $\bar{C}_B$. The top K BP observables were then selected to form:

$$\mathcal{B}_K = \{B_1, B_2, \dots, B_K\}.$$

This coherence-based ranking was used because the analysis aimed to evaluate how discriminability changes as increasingly broader sets of compact activity–width observables are included.

SM12.4 *K*-grid

The observation-lens analysis was performed across a fixed set of K values:

$$K \in \{20,\ 50,\ 100,\ 200,\ 500\}.$$

These values were selected to span small, intermediate, and broad observable-set scales.

Small K values represent highly selective observation using only the most coherent BP observables. Larger K values represent broader transcriptomic process coverage, but may also introduce redundancy, width burden, and less outcome-relevant signals.

SM12.5 K-dependent activity-magnitude term

For each selected observable set $\mathcal{B}_K$, the patient-level activity-magnitude term was recomputed as:

$$M_p^{(K)} = \frac{1}{K} \sum_{B \in \mathcal{B}_K} |\, \mu_{B,p} \,|.$$

This quantity measures the average activity magnitude of the top K selected BP observables in patient p.

SM12.6 *K*-dependent observable-width burden

For the same selected observable set $\mathcal{B}_K$, the patient-level observable-width burden was recomputed as:

$$U_p^{(K)} = \frac{1}{K} \sum_{B \in \mathcal{B}_K} \log\,(W_{B,p} + \delta).$$

This quantity measures the average log-transformed observable-width burden of the selected KBP observables in patient p.

SM12.7 *K*-dependent uncertainty-aware score

For each K and λ, the *K*-dependent uncertainty-aware patient-level score was defined as:

$$D_{W\text{-}PHY,p}^{(K)}(\lambda) = M_p^{(K)} - \lambda U_p^{(K)}.$$

Substituting the definitions of $M_p^{(K)}$and $U_p^{(K)}$:

$$D_{W\text{-}PHY,p}^{(K)}(\lambda) = \frac{1}{K} \sum_{B \in \mathcal{B}_K} |\, \mu_{B,p} \,| - \lambda \left[\frac{1}{K} \sum_{B \in \mathcal{B}_K} \log\,(W_{B,p} + \delta) \right].$$

This score was used to stratify patients for survival analysis at each observation scale K.

SM12.8 λ-scan within each K

For each K, the λ-scan described in SM9 was repeated over the fixed grid:

$$\lambda \in \{0,\ 0.25,\ 0.50,\ 0.75,\ 1.00,\ 1.50,\ 2.00\}.$$

For each pair (K, λ), patients were scored using:

$$D_{W\text{-}PHY,p}^{(K)}(\lambda),$$

stratified by median split, and evaluated using Kaplan–Meier and log-rank analysis. The corresponding clinical discriminability was:

$$D_{\text{clinical}}(K,\lambda) = -\log_{10}\big(p_{\text{logrank}}(K,\lambda)\big).$$

SM12.9 Scale-dependent discriminability curve

For each cancer cohort and each K, the best clinical discriminability over the tested λ-grid was defined as:

$$D(K) = \max_{\lambda}\ D_{\text{clinical}}(K,\lambda).$$

This generated a scale-dependent discriminability curve:

$$K \mapsto D(K).$$

The curve $D(K)$ was used to evaluate whether increasing the number of selected BP observables improved outcome separation monotonically or produced non-monotonic observation-lens behavior.

This directly supports Figure 5A of the main manuscript, where representative scale-dependent discriminability curves are shown across cancers.

SM12.10 Optimal observation scale

For each cancer cohort, the optimal tested observation scale was defined as:

$$K^* = \arg\max_K D(K).$$

The corresponding maximum discriminability was:

$$D_{\max} = \max_K D(K).$$

Because K^* was selected from a fixed grid, it should be interpreted as the best-performing tested observation scale rather than as a continuous estimate of true biological dimensionality.

This metric supports Figure 5B and Figure 5C of the main manuscript. Figure 5B summarizes cancer-specific optimal observation scales, while Figure 5C summarizes the maximum discriminability achieved at the optimal scale.

SM12.11 Interpretation of observation-lens effects

K-scale analysis was interpreted as an observation-lens sensitivity analysis.

If:

$$D(K)$$

increases with K, broader BP observable integration improves survival-based discriminability under the tested framework.

If:

$$D(K)$$

decreases with K, adding more BP observables may dilute coherent outcome-relevant signal.

If:

$$D(K)$$

peaks at an intermediate K, the result suggests an observation-lens effect: too few BP observables may miss relevant process-level information, whereas too many may introduce redundancy, diffuse width burden, or less relevant biological variation.

The main manuscript reports that uncertainty-aware discriminability showed non-monotonic dependence on K across cancers, supporting the view that observable quantity alone does not guarantee improved interpretability or clinical separation.

SM12.12 Relationship to Figure 5

SM12 directly supports Figure 5.

Figure 5A used representative curves:

$$D(K) = \max_{\lambda} D_{\text{clinical}}(K, \lambda)$$

to show scale-dependent discriminability across cancers.

Figure 5B summarized cancer-specific optimal observation scales:

$$K^* = \arg \max_{K} D(K).$$

Figure 5C summarized maximum discriminability achieved at optimal observation scale:

$$D_{\max} = \max_{K} D(K).$$

Together, these panels evaluate whether process-level discriminability depends on the number of selected BP observables and whether intermediate observation scales can outperform very small or broad observable sets.

SM12.13 Interpretation constraints

The *K*-scale analysis was not interpreted as a direct measurement of latent biological dimensionality, pathway complexity, or tumor-system size.

Instead, K was interpreted operationally as the number of BP observables included in the patient-level score. Therefore:

$$K^*$$

represents the best-performing tested observation scale under the present scoring, endpoint, gene-set collection, and survival-analysis framework.

It should not be interpreted as a universal cancer-specific biological constant.

Similarly, non-monotonic $D(K)$ behavior should be interpreted as evidence that observation design is scale-sensitive, not as proof of a specific mechanistic scale of tumor organization.

SM12.14 Output of *K*-scale analysis

For each cancer cohort, the K-scale workflow generated:

$$D_{\text{clinical}}(K, \lambda),$$

clinical discriminability for each K and λ;

$$D(K),$$

the best discriminability at each K;

$$K^*,$$

the optimal tested observation scale;

$$D_{\max},$$

the maximum scale-dependent discriminability; and the corresponding best-performing λ at each K, when required for summary tables.

These outputs were used to generate Figure 5 and supplementary observation-scale summaries.

SM12.15 Summary

In summary, SM12 defines the K-scale observation-lens analysis used to evaluate whether uncertainty-aware discriminability depends on the number of selected BP observables. BP observables were ranked by coherence, and the top Kobservables were selected for patient-level score construction. For each K, the uncertainty-aware score was computed as:

$$D_{W\text{-}PHY,p}^{(K)}(\lambda) = M_p^{(K)} - \lambda U_p^{(K)}.$$

Clinical discriminability was then evaluated across the λ-grid:

$$D_{\mathrm{clinical}}(K,\lambda) = -\log_{10}\big(p_{\mathrm{logrank}}(K,\lambda)\big),$$

and summarized as:

$$D(K) = \max_{\lambda} D_{\mathrm{clinical}}(K,\lambda).$$

The optimal tested observation scale was:

$$K^* = \arg\max_{K} D(K).$$

This analysis supports Figure 5 and shows that the usefulness of process-level transcriptomic observables depends not only on individual BP activity and width, but also on the observable-set scale used to construct patient-level discriminability scores.

SM13 | Supplementary figures, code organization, and reproducibility

All analyses in this study were implemented using reproducible Python-based workflows. The computational framework was organized to support transcriptomic preprocessing, biological-process gene-set mapping, patient-level activity and width construction, uncertainty-aware discriminability analysis, random-baseline validation, K-scale observation-lens analysis, and manuscript figure generation.

This section describes the output structure, figure-generation workflow, code organization, and reproducibility safeguards used to support the main and supplementary analyses.

SM13.1 Reproducibility objective

The reproducibility objective was to ensure that all major analyses reported in the manuscript could be regenerated from scripted workflows rather than manual figure editing or cohort-specific ad hoc processing.

The reproducibility framework was designed around four principles:

1. all cancer cohorts were processed using the same preprocessing logic;
2. all BP observables were constructed using the same patient-level $\mu_{B,p}$ and $W_{B,p}$ definitions;
3. all survival-based discriminability analyses used the same median-split, Kaplan–Meier, and log-rank procedures;
4. all figures were generated from exported summary tables using scripted figure-generation workflows.

This design was used to reduce hidden parameter tuning, manual intervention, and cancer-specific workflow bias.

SM13.2 Main computational workflow

The complete computational workflow followed the analysis sequence below:

TCGA expression and survival data
→ gene-expression preprocessing and normalization
→ BP and pathway gene-set mapping
→ patient-level $\mu_{B,p}$ and $W_{B,p}$ construction
→ cohort-level activity–width summaries

→ uncertainty-aware patient-level scoring

→ survival-based clinical discriminability

→ λ-scan analysis

→ random-baseline and D + coherence filtering

→ K-scale observation-lens analysis

→ cross-cancer integration and figure generation.

This workflow corresponds to the main manuscript structure, where Figure 2 reports λ-dependent discriminability, Figure 3 reports activity–width geometry, Figure 4 reports random-baseline and coherence diagnostics, and Figure 5 reports K-scale observation-lens effects.

SM13.3 Output objects generated by preprocessing

The preprocessing workflow generated the following cohort-level objects:

$$Z,$$

the cohort-wise z-score-normalized expression matrix;

$$G_{\text{expr}},$$

the cohort-specific gene universe;

$$P_{\text{matched}},$$

the matched patient set; and:

$$S = \{(p_i, t_i, e_i)\}_{i=1}^{N},$$

the survival table containing patient identifier, survival time, and event status.

These objects were used as the common input for BP mapping, patient-level observable construction, random gene-set generation, and survival-based discriminability analysis.

SM13.4 Output objects generated by BP observable construction

For each cancer cohort and gene-set database, BP observable construction generated:

$$B_{\mathrm{mapped}},$$

the mapped gene members for each BP observable;

$$N_B,$$

the mapped gene-set size;

$$\mu_{B,p},$$

the patient-level BP activity score;

$$W_{B,p},$$

the patient-level FWHM-defined observable width;

$$|\,\bar{\mu}\,|_B,$$

the cohort-level average activity magnitude;

$$\bar{W}_B,$$

the cohort-level average observable width; and:

$$\bar{C}_B,$$

the cohort-level coherence summary.

These outputs supported the construction of activity–width landscapes and coherence summaries in Figure 3.

SM13.5 Output objects generated by discriminability analyses

The uncertainty-aware scoring and survival-analysis workflow generated:

$$M_p,$$

the patient-level activity-magnitude term;

$$U_p,$$

the patient-level observable-width burden;

$$D_{W\text{-}PHY,p}(\lambda),$$

the patient-level uncertainty-aware stratification score;

$$D_{\mathrm{clinical}}(\lambda),$$

the survival-based clinical discriminability value;

$$\lambda^*,$$

the best-performing tested uncertainty-weighting value;

$$D_{\max},$$

the maximum λ-dependent discriminability;

$$\Delta D_{\text{baseline}},$$

the discriminability shift relative to $\lambda = 0$; and:

$$A_{\lambda},$$

the uncertainty-sensitivity amplitude.

These outputs supported the λ-scan results in Figure 2.

SM13.6 Output objects generated by random-baseline analysis

The random-baseline workflow generated:

$$\mathcal{D}_{\text{random}},$$

the empirical random discriminability distribution;

$$Q_{95}^{\text{random}},$$

the 95th percentile of random discriminability;

$$D_{\text{random}}^{\max},$$

the maximum random-set discriminability;

$$D_{\text{structured}}^{\max},$$

the maximum structured BP discriminability;

$$F_{\text{above}},$$

the fraction of structured BP observables exceeding random-background expectation;

$$F_{D+C},$$

the fraction of structured BP observables passing both random-background D and coherence filters; and:

$$J_{\max}(R),$$

the maximum Jaccard overlap between each high-D random set and structured BP annotations.

These outputs supported Figure 4A–D.

SM13.7 Output objects generated by K-scale analysis

The K-scale observation-lens workflow generated:

$$D_{\text{clinical}}(K, \lambda),$$

clinical discriminability for each K and λ;

$$D(K) = \max_{\lambda} D_{\text{clinical}}(K, \lambda),$$

the best discriminability at each observation scale;

$$K^*,$$

the best-performing tested observation scale; and:

$$D_{\max},$$

the maximum K-scale discriminability.

These outputs supported Figure 5A–C.

SM13.8 Main figure-to-method mapping

The main figures were supported by the following Supplementary Methods sections:

Figure 1

Conceptual and analytical workflow. Supported by SM1–SM6.

Figure 2

λ-dependent uncertainty-aware discriminability. Supported by SM7–SM9.

Figure 3

Cross-cancer activity–width geometry, database-level width summaries, coherence-based organization, and gene-set size dependence. Supported by SM3–SM6.

Figure 4

Random-baseline diagnostics, structured BP fraction above random background, structured/random maximum D comparison, D + coherence filtering, and biological-overlap analysis. Supported by SM10–SM11.

Figure 5

K-scale observation-lens analysis. Supported by SM12.

This mapping was used to ensure that each main-text figure had a corresponding methodological definition in the Supplementary Methods.

SM13.9 Supplementary figure generation

Supplementary figures were generated from exported summary tables and intermediate analysis outputs. These figures were designed to provide additional support for:

λ-scan sensitivity,
μ–W activity–width landscapes,
coherence distributions,
random-background behavior,
high-D random overlap diagnostics,
K-scale discriminability,

and:

cross-cancer summary behavior.

Supplementary figures were generated programmatically using Python plotting workflows. No manual data manipulation was required for figure regeneration after the summary tables were exported.

SM13.10 Code organization

The code repository was organized into modular workflow components corresponding to the major analysis layers:

Preprocessing scripts

Used for expression matrix loading, gene-symbol standardization, TCGA barcode harmonization, survival-table parsing, sample matching, primary tumor filtering, and z-score normalization.

Gene-set mapping scripts

Used for GMT parsing, gene-set standardization, mapped gene filtering, and construction of retained BP observable lists.

Observable construction scripts

Used for patient-level $\mu_{B,p}$, patient-level $W_{B,p}$, cohort-level activity–width summaries, and coherence calculation.

Discriminability scripts

Used for $D_{W\text{-}PHY,p}(\lambda)$, median-split stratification, Kaplan–Meier analysis, log-rank testing, λ-scan analysis, and clinical discriminability calculation.

Random-baseline scripts

Used for size-matched random gene-set generation, random D distributions, random 95th percentile thresholding, structured/random comparison, D + coherence filtering, and high-D random overlap analysis.

K-scale scripts

Used for coherence-based BP ranking, top-K BP observable selection, K-dependent scoring, $D(K)$construction, and K^*identification.

Figure-generation scripts

Used for loading exported CSV summaries and rendering main and supplementary figures.

SM13.11 Repository documentation

To support reproducibility and workflow traceability, the repository includes organizational documents such as:

README.md,

RUN_ORDER.md,

FIGURE_CODE_MAP.md,

and:

SUPPLEMENTARY_METHODS_CODE_MAP.md.

The README file describes the purpose and organization of the repository. RUN_ORDER describes the recommended execution sequence. FIGURE_CODE_MAP links figure panels to figure-generation scripts and input summary tables. SUPPLEMENTARY_METHODS_CODE_MAP links Supplementary Methods sections to the corresponding computational modules.

These files were included to make the relationship between manuscript methods, analysis scripts, and output figures explicit.

SM13.12 Figure-regeneration protocol

The figure-regeneration workflow followed a two-stage structure.

First, analysis scripts generated intermediate and summary outputs, including:

μ/W summary tables,

λ-scan tables,

random-baseline tables,

$D + C$ filter tables,

$Jaccard$ overlap tables,

and:

K-scale summary tables.

Second, figure-generation scripts loaded these exported tables and regenerated figure panels.

This separation allowed figures to be regenerated without rerunning the full transcriptomic preprocessing and survival-analysis workflow, provided the summary CSV files were available.

SM13.13 Data availability

The transcriptomic and clinical datasets analyzed in this study were obtained from publicly available TCGA resources through the UCSC Xena platform. The main manuscript also states that the datasets are publicly available from TCGA through UCSC Xena.

Gene-set collections were obtained from publicly available GMT-compatible pathway resources, including Hallmark, GO:BP, and Reactome collections.

Processed summary tables generated during analysis were used for figure construction and downstream reproducibility checks.

SM13.14 Code availability

The code used for transcriptomic preprocessing, biological-process mapping, observable-geometry construction, uncertainty-aware discriminability analysis, structured-versus-random validation, and supplementary figure generation is available at the GitHub repository listed in the main manuscript. The main text provides the repository location under Code Availability.

The repository is intended to support independent rerunning of the analysis workflow and regeneration of manuscript figures from exported summary files.

SM13.15 Reproducibility safeguards

Several safeguards were used to improve computational reproducibility:

1. cohort-wise preprocessing was performed using the same workflow across cancer types;
2. no pooled cross-cancer normalization was used;
3. gene-set mapping used fixed mapped-gene thresholds;
4. $W_{B,p}$ was defined consistently as patient-level within-BP FWHM;
5. $\delta = 10^{-6}$ was used for numerical stabilization;
6. the same λ-grid was applied across cohorts;
7. the same K-grid was applied for observation-lens analysis;
8. random gene sets were size-matched by N_B;
9. survival analysis used the same median-split and log-rank procedures;
10. figures were generated from exported summary tables.

These safeguards were designed to reduce hidden tuning and ensure that differences across cancer cohorts reflected differences in observable structure rather than workflow inconsistency.

SM13.16 Computational limitations

Several computational limitations should be considered when interpreting the results.

First, FWHM estimation may be unstable for very small or highly discrete gene-expression distributions. This limitation was reduced by mapped-gene filtering and fallback width estimation but cannot be fully eliminated.

Second, survival-based discriminability depends on cohort size, event number, endpoint quality, and censoring structure. Cancers with sparse events may produce less stable D_{clinical} estimates.

Third, random-baseline analysis depends on the number of random repetitions and the available transcriptomic gene universe. Random controls were used as empirical diagnostic references rather than definitive null distributions.

Fourth, K^* and λ^* were selected from fixed tested grids and should not be interpreted as continuous optimized biological constants.

Fifth, the analysis was based on bulk gene-expression data and did not directly incorporate spatial, temporal, single-cell, proteomic, perturbational, or network-topology information.

SM13.17 Summary

In summary, SM13 defines the reproducibility framework for the D-PHY-I analysis. The workflow was implemented using modular Python scripts covering TCGA preprocessing, BP gene-set mapping, patient-level activity and width construction, uncertainty-aware scoring, survival-based discriminability, random-baseline validation, D + coherence filtering, biological-overlap diagnostics, K-scale observation-lens analysis, and figure generation.

The main reproducibility structure can be summarized as:

$$\text{raw data} \rightarrow \text{processed expression and survival objects} \rightarrow \mu_{B,p}, W_{B,p}$$
$$\rightarrow \text{summary CSV files} \rightarrow \text{main and supplementary figures.}$$

This organization was used to ensure that the analytical workflow, figure generation, and manuscript interpretation were traceable from data preprocessing to final outputs.

Several limitations apply to the Supplementary Methods. First, $W_{B,p}$ is an operational width estimate and does not decompose biological heterogeneity, projection ambiguity, and technical variability. Second, survival-based D_{clinical} depends on endpoint quality, event number, censoring structure, and cohort size. Third, λ^* and K^* are selected from fixed tested grids and should not be interpreted as continuous biological constants. Fourth, random-baseline analyses provide empirical diagnostic references rather than definitive mechanistic null models.